

Automated Micro pipetting System for Users with Limited Dexterity

Final Report



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Revision	Date	Author	Description
A	18/06/2025	Group F	Final release

18 June 2025

Word Count: 4986

Abstract

Manual Micro pipetting is essential in wet lab environments but presents significant challenges for individuals with tremors, fine motor impairments or limited dexterity. This accessibility barrier limits participation and independence in scientific research and lab work. This project addresses this gap by developing a compact, automated pipetting system to enable independent use by individuals suffering from physical disabilities. This system enables accurate, repeatable pipetting without the need for sustained fine motor control, whilst achieving volume delivery within $\pm 1\%$ of target volume.

The system integrates a 3-axis Gantry system, linear actuator and touchscreen/keypad interface to eliminate the requirement of precise manual dexterity. By mechanising the liquid aspiration and dispensing processes, this design helps to reduce human error and physical strain, while maintaining required precision for laboratory protocols. The core objectives were to increase independence of disabled users, reduce reliance on assistance in laboratory environments, and maintain the accuracy of traditional pipetting.

Development priorities include ergonomics, mechanical stability, a comprehensive user interface, and consistent volume control. A compact design and lightweight construction ensure the device is portable, even for users with reduced strength or reach. The device accommodates standard pipette tips and volumes and is operable via simple interface elements such as a numerical keypad or touchscreen inputs. Initial research involving Imperial College Disability Services, informed the design and validated the systems usability and ability to improve accessibility.

In conclusion, the Automated Micro pipetting System demonstrates a viable approach to enhancing accessibility within laboratory environments. It offers a scalable solution for increasing inclusion and accessibility in research and education without compromising scientific performance.

Future developments may incorporate programmable volume sequences and mobile app integration to help further streamline workflows and enhance ease of use.

Acknowledgements

Our sincerest thanks go to Ian Radcliffe, our project supervisor, whose expert oversight and thoughtful guidance was invaluable throughout the development process. His input on design feasibility and manufacturing process played a key role in refining our final design.

We are also grateful to the Imperial Disability Advisory Services and Dr Katherine Dean for their contributions during the initial research phase. Their input helped shape our project's direction by highlighting key accessibility considerations and guiding our early design decisions.

We appreciate the assistance of Ms Iris Latham - our disability consultant. Her insight on the unique experiences of people with limited motor skills allowed us to make the necessary improvements for a more accessible and inclusive design.

A special thanks goes to Ms Mandy Mak, Mr Paschal Egan, and Mr Tariq Malik for their constructive feedback during the design phase. Their insights prompted significant improvements, making our final design more practical and effective.

Lastly, we'd like to thank Dr Tweety Tang for organising and facilitating user feedback sessions to ensure our project aligned with our group mission to improve accessibility.

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1 Introduction

Manual micro pipetting is a core technique in biological and chemical laboratories. It is essential for obtaining accurate experimental results. However, individuals with fine motor impairments, such as tremors, reduced dexterity, or limited upper limb mobility, often face significant challenges in performing these tasks. For some, pipetting may be time-consuming, frustrating, or entirely inaccessible without assistance. This creates a physical and professional barrier, limiting both independence in laboratory work and equal participation in scientific research and education.

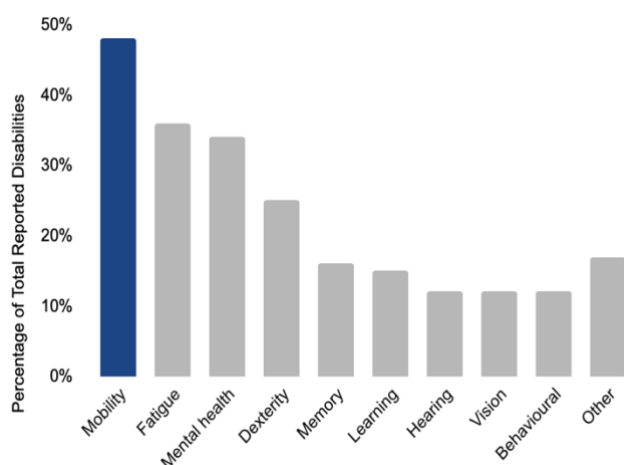
To address this critical accessibility gap, this project focused on the design and development of a low-cost, compact, automated micropipetting system. The goal was to create a device that enables users with physical impairments to independently and accurately perform pipetting tasks, thereby promoting inclusivity in laboratory environments.

Rather than targeting a single type of disability, the project was designed with a broad spectrum of upper body impairments in mind. This inclusive approach maximizes the potential impact of the device. The system is intended to assist individuals with tremors, who may struggle to hold even a beaker steadily; those with fatigue, who may find repetitive pipetting tasks exhausting; and users with limited mobility in one arm, who may be unable to operate a manual pipette but can interact with a touchscreen or interface, etc.

1.1 Background Research

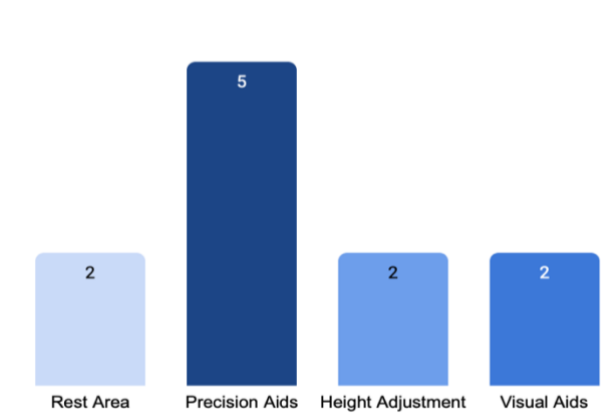
Prior to selecting a pipetting system as the focus, the team conducted research into the broader challenges faced by individuals with disabilities in laboratory settings. This included reviewing academic literature, exploring public data on disability prevalence, and conducting interviews with individuals in STEM societies, educators, and disabled individuals with laboratory experience.

Figure 1: Prevalence of Disability Types in the UK (%)



Note: Respondents often reported multiple disabilities; totals exceed 100% [1]

Figure 2: Key Accessibility Challenges Reported by Disabled Users



Note: Respondents were contacted through Royal Society of Chemistry, Imperial College disability services, etc

Upper body impairments and tasks requiring fine precision emerged as common themes. According to the UK House of Commons, Figure 1, mobility-related issues, including upper body mobility, are the most prevalent form of disability, with nearly 50% of disabled individuals reporting such challenges. Fatigue and dexterity were also highly reported, which our product directly targets. In our independent outreach, figure 2, participants frequently

cited difficulties with fine motor tasks, reinforcing the need for solutions that support precision work.

The team also evaluated existing adaptive technologies in laboratory environments. While tools such as tactile microscopes, talking scales, and adjustable lab benches exist [2] [3] [4]; few solutions specifically support fine motor tasks for users with upper limb impairments. Commercial liquid-handling robots, like those from Tecan [5] and SPT Labtech [6], do offer automation but are typically expensive, complex, and designed for high-throughput industrial or research applications. They are unsuitable for day-to-day use by individuals requiring personal assistance in smaller academic or teaching laboratories. Moreover, they are rarely designed with accessibility for disabled users in mind.

This project seeks to bridge the gap between accessibility and precision by offering a compact, cost-effective, and user-friendly automated pipetting solution. Designed to integrate into standard lab protocols and compatible with common pipette tips and volumes, the system uses intuitive controls that require minimal manual dexterity. The resulting device supports independent operation by users with physical impairments, without compromising the accuracy essential to scientific research.

2 Requirements Definition

The Automated Micro Pipetting System was designed specifically with end users in mind, aiming to support individuals with upper body impairments in performing pipetting tasks independently and accurately. The key user requirements reflect the need for improved accessibility, usability, and precision, whilst ensuring seamless integration into standard laboratory environments and remaining intuitive and easy to use for those with limited dexterity.

Figure 3: Primary User Requirements

Clear Feedback

The system should provide real time feedback (visual) to notify users about current processes

Intuitive design

Device's interface and controls must be intuitive, requiring minimal training or prior knowledge

Compact Design

The device must fit on typical laboratory bench without obstructing other required laboratory equipment

Simple Operation

The system's controls and interaction methods should be clear and easy to use for individuals with a range of mobility disabilities

Minimal Fine Motor Input

Control mechanisms should avoid requiring precise movements or sustained grip strength

Standard Lab Compatibility

Control mechanisms should avoid requiring precise movements or sustained grip strength.

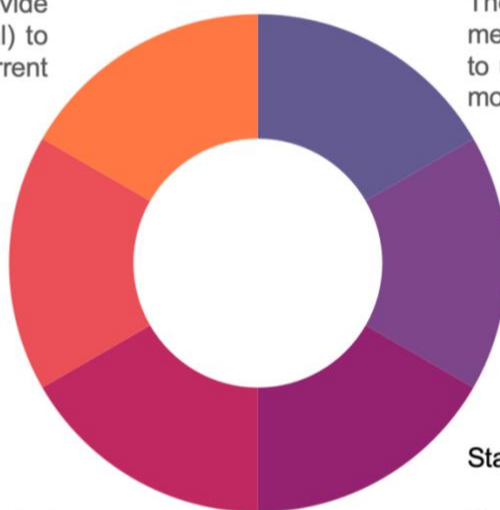
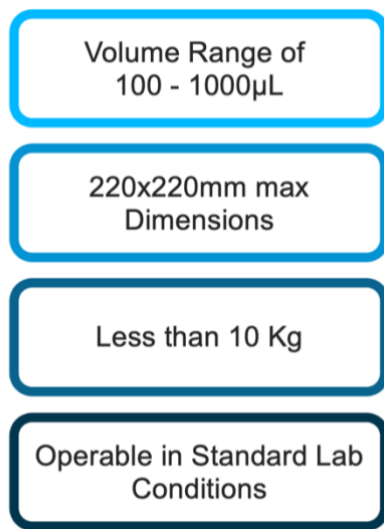


Figure 4: Key Quantitative Requirements



Pipetting Volume Range: 100µL – 1000 µL, with $\pm 1\%$ volume accuracy (adhering to ISO standards [8]).

System Footprint: 220mm x 220mm maximum dimensions, to ensure laboratory bench compatibility [7]

Lightweight Design: Less than 10 kg to ensure it can be comfortably lifted and relocated to accommodate disabled users

Operable within Laboratory Settings: Should be operable in standard laboratory conditions (e.g. temperature in range 18-22°C [9])

Sources: [7] [8] [9]

The requirements outlined in Figure 3 & 4 guided the electronic and mechanical design decisions, as well as the testing and evaluation strategies. They also serve as a benchmark to ensure the system's success for meeting user needs and ensuring that accessibility enhancements don't come at the expense of laboratory efficiency or accuracy.

3 Final Design

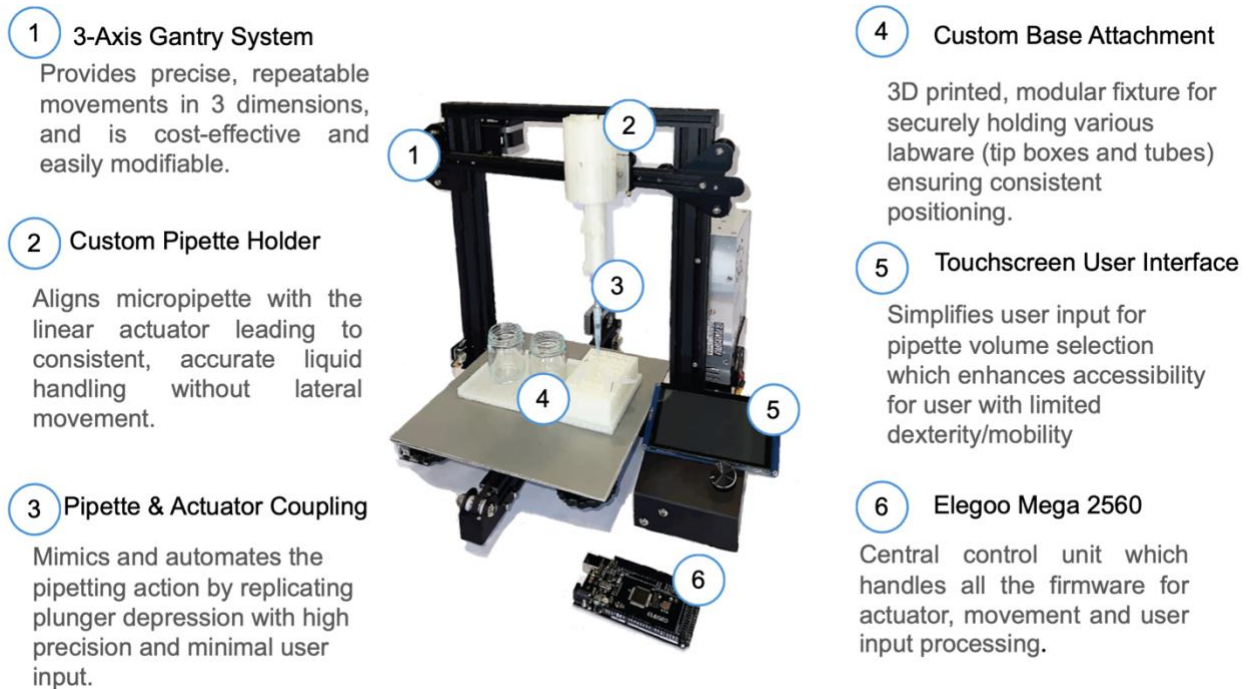
3.1 Assembly and Overview

The final design, which is outlined in Figure 5, is a compact, automated micro pipetting device specifically engineered to assist users with limited hand dexterity in performing accurate and independent liquid handling tasks. The system integrates precise and direct movements in 3 dimensions using stepper motors mounted on a modified 3D printer gantry system as a base (adapted from Ender 3 3D printer) [10]. This foundation provided a lower cost, robust mechanical platform and reduced the need for custom design and manufacturing. The pipette mechanism is operable through a user-friendly LCD touchscreen interface, allowing programmable volume control, live feedback, and intuitive operation.

At the core of the pipetting mechanism is a custom designed 3-D printed pipette holder, designed to securely mount and operate an open-source 3D printed Adjustable Micropipette which complies with the ISO standards for accuracy [7]. This pipette can be operated mechanically by the system via an actuator to aspirate and dispense fluids with programmable volume control, set via the user-friendly interface. The user selects desired volume, and the microcontroller executes the movement sequence accordingly.

The design also includes a modular base component equipped to hold commonly used laboratory equipment such as pipette tip boxes, microcentrifuge tube racks and sharps containers. This module improves accessibility and enhances workflow within standard laboratory environments. The entire system has been designed to be compatible with standard micropipette tips and corresponding volume ranges, ensuring it integrates smoothly within existing laboratory setups.

Figure 5: Final Design Features of Automated micro-Pipetting System



Note: Final design also includes electronic components not visible in diagram, such as LAC board, motors, motor drivers, buck converters and the linear Actuator

3.2 Gantry System

The gantry system forms the structural backbone of the device, enabling precise, repeatable movements in X, Y and Z planes. It is built upon the assembly of a Creality Ender 3D printer [10], selected for its affordability, structural rigidity and pre-integrated movement components. This includes linear rails, mechanical end-stop switches and lead screws allowing for smooth, accurate translational movements. The gantry system was calibrated by determining the number of stepper motor pulses required to translate a specific linear distance, allowing accurate mapping between step count and physical displacement along the axes. Empirical calculation showed us that there was a movement of 0.2mm per step in the Y and X direction, and a movement of 0.04mm per step in the Z direction. (Appendix I)

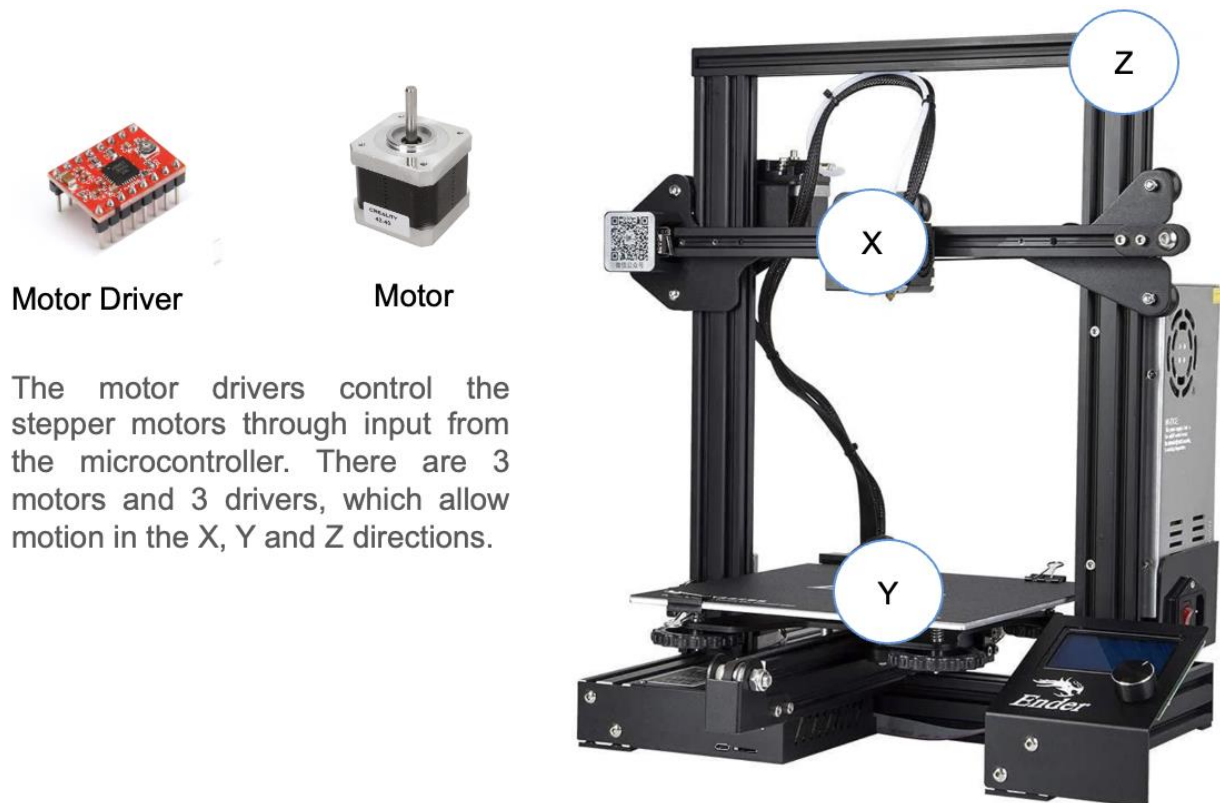
To adapt this system, we replaced the stock stepper motor drivers with A4988 stepper motor drivers (figure 6), interfaced via a complementary circuit connected to a microcontroller (Elegoo Mega 2560). This modification ensures full integration with the custom electronics (Appendix F) and firmware, enabling precise pipette positioning through programmable control inputs. The travel limits are approximately 220mm in X and 220mm in Y, and 170mm in the Z dimension. Mechanical end-stops were retained along all the axes to enable consistent mapping of positions from startup, ensuring positional accuracy.

The configuration is as follows:

- X-axis: lateral movement of the pipette, allowing precise positioning across.
- Y-axis: forwards and backwards translation of base platform.
- Z-axis: Vertical movement to raise/lower the pipette for aspiration.

By moving the base platform into position, the system enables reproducible and automated positioning specific to liquid handling tasks being performed.

Figure 6: Elements of the Gantry System

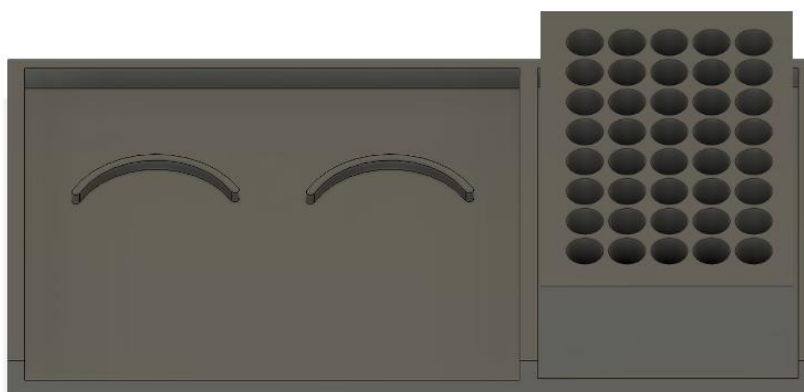


Source: [10] [11] [12]

3.3 Base Attachment

The base attachment was designed in SolidWorks (Figure 7) to slot onto and interface seamlessly with the gantry system. It acts as the main stage for labware and sample containers, housing secure slots for pipette tip boxes, microcentrifuge tubes, and other modular lab equipment. The positions of these slots are predefined on the plate, allowing for consistent accurate positioning throughout the automated pipetting process.

Figure 7: Base and Cuvette Holder CAD



Note: Renderings of our custom designed microcentrifuge tube holder (left) and modular base attachment (right), both developed to ensure stable positioning of labware during automated pipetting processes

The base is securely affixed onto the printer bed, allowing for controlled movement in 3 dimensions via the Gantry System. This allows for precise translation of the labware elements directly under the pipette for tasks such as fluid dispensing or tip pickup.

This attachment is 3D printed from PLA, chosen for its rigidity, ease of printing, and mild chemical resistance [13] [14] [15]. Labware slots are custom designed to accommodate standard containers, for example the microcentrifuge tube holder and standard beaker sizes [16], ensuring easy slotting and stable positioning during dynamic motion. The integrated locating grooves also serve as guard rails, providing additional physical guidance to allow secure placement of labware even during dynamic motion, and assisting users with reduced hand dexterity by simplifying the loading process.

3.4 3D-Printed Adjustable Micropipette

The liquid handling system incorporates an open-source, 3D printable adjustable micropipette designed to meet ISO 8655 standards, representing accuracy, precision and reliability [17]. This pipette operates by actuating a disposable tip design syringe to a user specified point, enabling precise volume control without the necessity of complicated internal mechanisms. Figure 8 shows the sequence of events involved in pipette usage, demonstrating how the plunger displacement leads to liquid aspiration and dispensing.

Figure 8: Sequence of events involved in pipette usage



Note: Shows CAD modelling of operating modes. A - Latched position ready to aspirate, after the button is pressed releasing pipette to position B. B - Set position, screw position determines the displacement of plunger and therefore aspiration volume. C - The blowout position, plunger is pressed in dispensing liquid. D - Return to the latched position, ready to repeat process. [17]

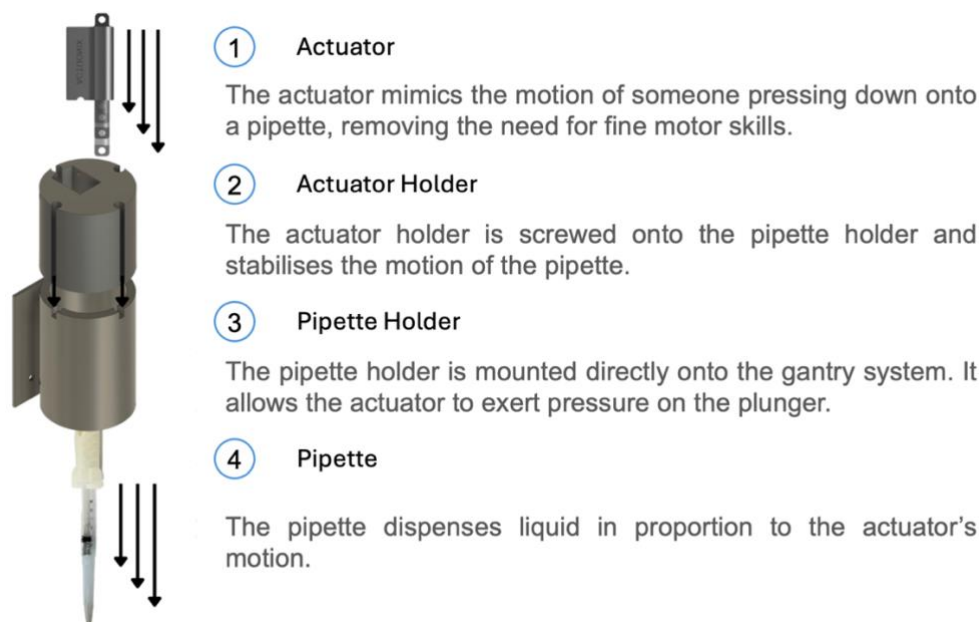
The micropipette is composed of 3D-printed components fabricated using FDM with PLA filament, combined with readily available hardware such as disposable syringes and springs. The design utilises the evenly spaced graduations on the syringe barrel to set the desired volume, allowing for straightforward adjustment and calibrations.

The pipette is configured to 100-1000 μL , utilising a 3ml syringe, volume range allowing us to adhere to our project specifications.

Initial testing from the design process revealed that the syringe's original graduation marks resulted in systematic error due to air compressibility effects [17]. To address this as part of the design we used, adjusted scale graduations were developed and affixed to the syringe, compensating for the accumulated error. With these adjustments, the pipette achieved compliance with ISO 8655, demonstrating accuracy and precision comparable to commercial micropipettes.

3.5 Actuator and Pipetting Mounting

Figure 8: Pipette Mechanism



Note: Holders are 3D renderings of our CAD models, while pipette is an image of our actual product

The actuator is directly mounted onto the X-axis of the gantry system using a custom designed 3D printed attachment, which securely holds both the actuator and the micropipette in precise alignment with one another (Figure 8). This setup ensures that the actuator's action is consistently transmitted to the pipette's plunger, ensuring repeatable and controlled aspiration/dispensing. The actuator's extending arm is mechanically joined to the plunger, essentially replicating the manual action performed while pipetting.

The holder is designed to accommodate our specific 3D printed pipette, providing a secure fit with easy removal and insertion. Features were specially designed for alignment of the pipette and specific clamping mechanisms were introduced to minimise lateral movement and ensure repeatability, critical for maintaining volumetric precision. The bracket is printed from PLA, which offers sufficient rigidity for repeated use while being easy to manufacture using FDM 3D printing. It is directly attached to the gantry's X-axis carriage using existing mounting holes, avoiding modifications to the pre-existing 3D printing frame helping maintain mechanical stability during operations provided by the 3D printer kit.

To control the actuator, a Linear Actuator Control (LAC) board is used, allowing for communication between the microcontroller and actuator. The actuator operates at 12 V, which is supplied by an external DC power source, while the control signals are provided via the microcontroller. The actuator's position is controlled through a PWM signal generated by the microcontroller. The LAC board then interprets these signals and modulates actuator extension accordingly by converting the pulse width into a proportional output voltage to drive the actuator. (Connection can be seen in Appendix F) [18]

User input is provided via the touchscreen interface/pin pad, where the desired pipetting volume is selected. The software maps a selected volume to a corresponding actuator position based on pre-established calibration data [17]. This ensures accurate displacement of pipette plunger, enabling the system to achieve volume delivery within the ISO 8655 [7] standards for accuracy ($\pm 1\%$ of desired volume). The use of a 3D-printed mount and direct mechanical connection minimises movement of individual parts and ensures stability during actuation, critical for maintaining pipette precision when running multiple cycles.

3.6 LCD Screen and User Interface

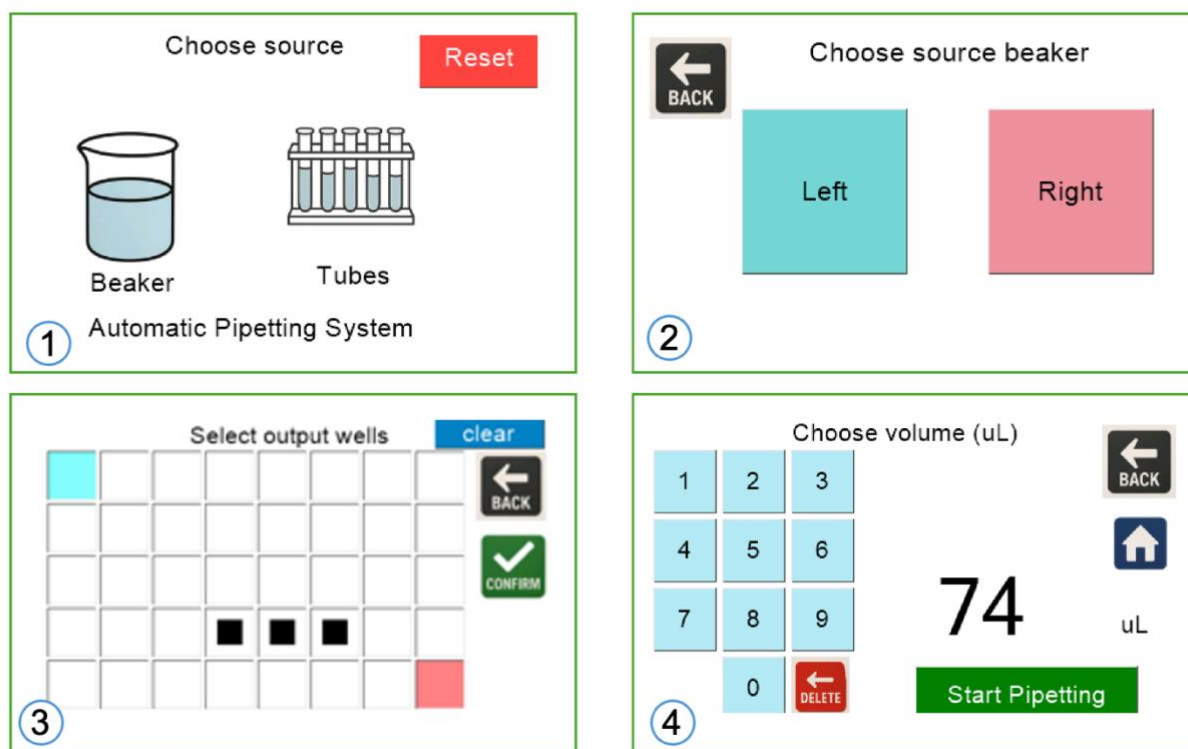
The user interface is composed of two main I/O components: a touchscreen display and a numerical keypad. The touchscreen display is integrated with the microcontroller via a UART serial connection, utilizing the TX (transmission) and RX (reception) pins in the Elegoo Mega 2560 for data transfer (Appendix F). The display is 5 V, so it is powered through a regulated power source from the microcontroller to ensure a consistent power supply. The screen serves as the primary output device, providing real time feedback to the user when they are selecting volumes and specific output wells.

For user input, a number pad is connected to the microcontroller using digital logic pins. The keypad enables manual volume entry and operational commands, offering an alternative input method in scenarios where touchscreen interaction is less practical, like if the user wants tactile feedback with the inputs. Each key press is mapped to specific functions within the software, allowing the user to select operations and volumes for specific tasks (aspiration/dispensing).

Integration of both the touchscreen display and the physical keypad enhances accessibility and flexibility of our system, catering to users with varying levels of physical ability. Direct digital connections to the microcontroller and regulated 5 V power rails maintain signal integrity throughout the system, ensuring reliable and consistent communication and operation of user interface components even within a laboratory setting.

Figure 9 shows the screen sequence that the user must input. Colours have been implemented in accordance with accessibility requirements. [19] A full explanation can be found in Appendix H.

Figure 9: Screen Layout images



Note: User interface of the Automatic Pipetting System showing the step-by-step process: (1) selecting the source type (beaker or tubes), (2) choosing the source beaker (left or right), (3) selecting a single input well from a grid, and (4) setting the pipetting volume before starting the operation.

3.7 Software

The firmware is written in C++ and compiled in the Arduino IDE, providing low level control over the system's mechanical and user interface components. The firmware implements a cartesian co-ordinate system, enabling precise, absolute position control across the axes.

Upon startup, a homing routine is automatically executed, where each axis head towards mechanical end stop switches, establishing a reference zero position which is critical for accurate and repeatable operations.

For coordinated movements in the X-Y plane we use synchronized stepper motor pulses and linear trajectories, which are crucial for targeting wells and labware. The Z axis operates independently, enabling vertical pipetting actions without disrupting X-Y positions. The motion speed is controlled via adjustable microsecond step delays; these are tuned specifically for each axis to balance smooth motion and operational efficiency.

The touchscreen interface was developed using Nextion, a graphical tool for designing embedded user interfaces. While the graphical layer is designed via the software, the underlying control logic is written in a C-like language.

User selections on the screen trigger the transmission of control variables to the Arduino. These inputs are parsed to generate corresponding motor sequences, enabling the pipette's motion based on user commands.

Safety features are implemented directly in the software, preventing axis movement beyond physical limits and automatically terminating operations in the event of a homing failure. Additional safeguards are built into the touchscreen interface, such as restricting user input to valid volume ranges and preventing entries that are too large or too small.

The modular nature of the code allows for future extensions, including Bluetooth Low Energy modules, enabling system control via mobile applications, further enhancing accessibility for users with limited reach or dexterity. A full version of the code can be viewed in Appendix G.

3.8 Viability

There is strong evidence supporting the viability of this project, based on both market trends and user needs. Similar automated pipetting products already exist, indicating a proven market [20]; however, these are typically expensive and aimed at experienced researchers working in high-throughput laboratory environments. In contrast, our product is designed for a niche but underserved demographic: students and researchers with disabilities in STEM education. By offering a significantly lower-cost and more accessible solution, we address a key gap in the market. Our bill of materials in Appendix D highlights that the total material cost of one unit would be £483.53, this would allow us to be very competitive with systems that can cost upwards of £35 000 [5].

Currently, most laboratory accessibility tools focus on general-purpose accommodations rather than direct interaction with lab instruments. For example, products such as accessible lab benches, voice-activated lab software, and adjustable-height equipment aim to improve physical access to the workspace. Tools like tactile lab aids and screen-reader-compatible interfaces have also been developed to support visually impaired students [2] [3]. However, there are very few specialized tools that support fine-motor tasks like pipetting, which are often critical in chemistry and biology labs. The few assistive pipetting devices that do exist,

such as ergonomic pipette aids or semi-automated dispensers, are generally designed for professionals and are not tailored for affordability or ease of learning.

Our solution is uniquely positioned to meet this gap by combining automation, low cost, and an education-focused user experience. Recent increases in the number of STEM students with disabilities, along with broader improvements in inclusivity and assistive technology in education, further support the relevance and potential popularity of our device [21]. Our approach combines affordability with a clear educational benefit, making it well-positioned for institutional adoption.

3.9 Scalability

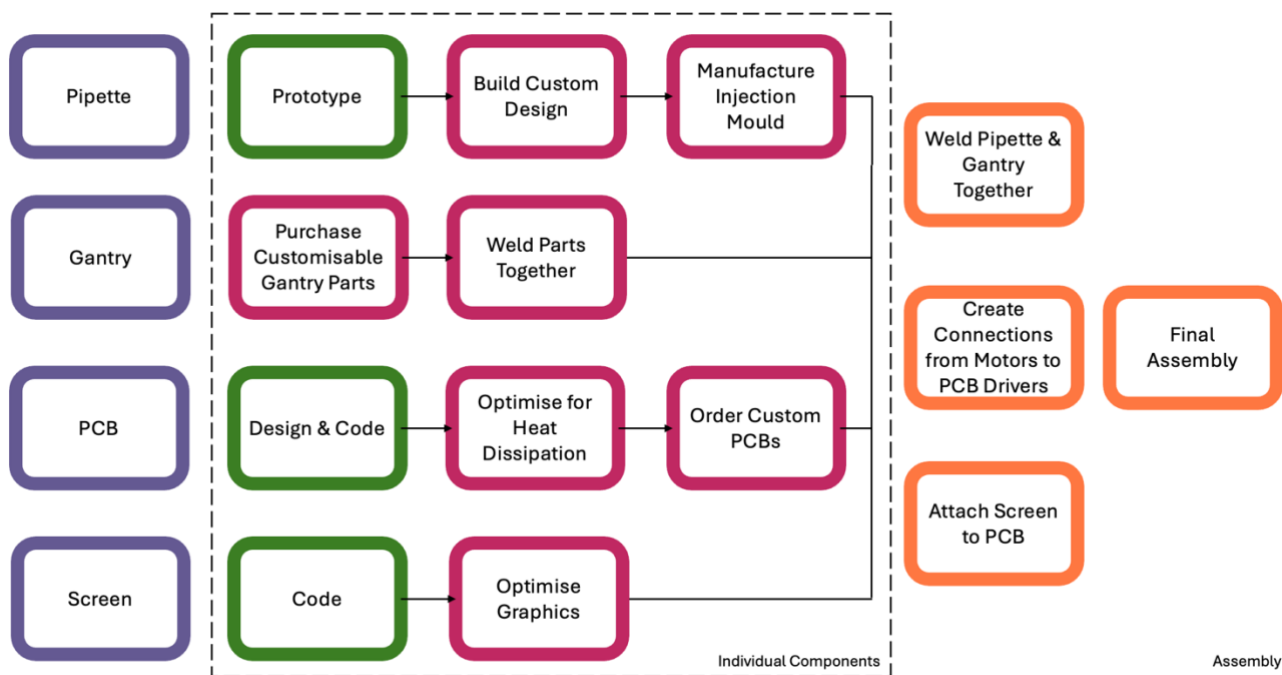
The final design has been developed with scalability in mind, ensuring a smooth transition from prototype to mass production with minimal modifications. At its core, the system utilizes a commercially available 3D printer platform, which is already mass-produced and widely accessible—significantly reducing the complexity of scaling up. Custom components, including the redesigned pipette and associated electronics, will require separate manufacturing; however, both are suitable for scalable mass production through methods such as injection moulding, CNC machining, and automated PCB assembly.

One current limitation lies in the attachment point between the pipette and the printer arm, which lacks sufficient stability in its current iteration. Replacing this joint with welded metal components will enhance durability and ensure long-term reliability.

To ensure a successful transition to full-scale production and deployment, further work is needed to identify and mitigate potential risks. These include mechanical failures under continuous use, variability in component quality across batches, electronic reliability, and user safety considerations in educational settings. Addressing these risks early through rigorous testing, quality assurance protocols, and design refinements will be essential to maintaining system performance and user confidence at scale. Risks can be viewed in Appendix B.

Nevertheless, the system's modular design supports straightforward scaling with only minor adjustments, making it a viable solution for broad educational deployment. Figure 10 outlines the remaining steps required before initiating mass production.

Figure 10: Flow chart representing manufacturing process



3.9.1 Electronics

The electronics are centred around a custom-designed PCB that integrates directly with the existing power and control systems of the 3D printer. While the current version is functional, there are opportunities to improve the layout, particularly by optimizing spacing between components. These adjustments would enhance heat dissipation, simplify servicing, and reduce the risk of interference during operation. From a manufacturing standpoint, the PCB is highly scalable and compatible with industry-standard processes such as PCB roller coating and automated pick-and-place assembly. For commercial deployment, the electronics must also comply with international standards. Specifically, the design must adhere to IEC 60446 for colour coding and identification and obtain CE certification and UKCA marking to meet regulatory requirements for safety and distribution [22] [23] [24].

3.9.2 Pipette & Holder

The current pipette is a functional prototype and has proven effective for initial testing. However, its 3D-printed construction presents significant challenges for large-scale production. The process is too time-consuming, and the resulting parts lack the durability and consistency required for reliable use in an educational setting. To address this, the pipette should be redesigned with manufacturability in mind. Potential approaches include using injection moulding or CNC machining to produce higher-quality components more efficiently. This redesign also presents an opportunity to improve the mechanical integration with the printer arm, potentially eliminating the need for a separate holder and simplifying the overall assembly.

3.9.3 Screen

The screen is coded using our own custom design. There are no issues with scaling the software or with mass producing the screen itself.

3.9.4 Gantry

The gantry has been repurposed from the 3d printer and should easily be scalable. There are certain design elements that may need to be slightly modified to fit the final design, such as the Z-axis limit switch arm.

4 Discussion

4.1 Design Evaluation

The automated micro pipetting system successfully fulfilled its core design requirement: allowing users with lacking fine motor control to successfully perform pipetting tasks independently and with reliable accuracy.

4.1.1 Evaluation Matrix

Requirements	Outcome	Evaluation	PSD [8] number reference
Accurately dispense between 100µL and 1mL within $\pm 1\%$ of the target volume	Able to calibrate pipette alone to complete this successfully, however actuator adds in extra uncertainty.	Partial success	1.1
Chemical compatibility and safety with solutions such as buffer, salt and biological reagents.	All individual components, such as PLA filament, should be moderately chemical resistant. However, there have been no tests for extensive exposure.	Partial success	1.2, 6.1, 9.3
Splash free dispensing due to controlled speed.	The actuator provides a controlled speed of liquid dispensing.	Success	1.3
Contamination free use.	Not contaminated by coloured liquids between use, however chemical contamination between uses was not tested.	Partial success	1.4
Simple operation for users of varying abilities.	According to user feedback 55% of users indicated that minimal instructions were necessary.	Success	1.5, 3.1
Bluetooth control.	Due to feedback received it was determined that an integrated touchscreen/ touchpad would be more accessible for limited mobility.	Not successful	1.6

Fit within lab bench of 75cm x 120-150cm.	The device is the able to fit on the lab bench, being 44cm x 44cm x 46.5cm.	Success	2.1
Weight under 10kg in order to allow transportation.	The system is under 10kg and according to user feedback can be moved by one to two individuals depending on mobility limits.	Success	2.4
Provides an adequate level of visual and auditory feedback.	The controls provide a level of visual feedback for users and minimal auditory feedback at the beginning and end of pipetting. (buzzer)	Success	3.2
Minimises physical strain while both sitting and standing.	User feedback indicated that this minimises physical strain induced by using a manual micropipette.	Success	3.3
Power consumption is compatible with mains power supply.	The power consumption of 24 watts is compatible with mains power supply due to a built-in adaptor on the gantry system.	Success	4.2
Easy maintenance with detailed instructions and available parts.	Parts were sourced locally, where possible and instructions are provided on how to use the system.	Success	7.1
Affordability should be under £600 to compare with similar equipment.	The total cost of the product was £483.53. This is well below the £600 target and can be optimised further with mass production (Appendix D)	Success	8.1
Accessible for as wide a range of users as possible.	Depending on the user's disability this product does provide a more accessible solution to micro pipetting according to user feedback.	Partial Success	9.5

Table 1. Table of the Evaluation of the requirements against the PSD.

4.1.2 Testing

The pipetting system was tested for both volumetric accuracy and cross-contamination. To assess accuracy, the system was programmed to dispense target volumes ranging from 30 to 100 μ L. The dispensed volumes from the automatic pipetting system were quantitatively compared to the dispensed volumes from a manual micropipette. It was observed that as volumes decreases, accuracy and precision decreased as volumes became more difficult to dispense [25].

Across the tested range, the system consistently achieve accuracy within a +/- 1% tolerance. Cross contamination was evaluated qualitatively. Between dispense cycles, dyed liquids were first run through the system, after each cycle the pipette tip was replaced, and a clear liquid was dispensed. There were no visual traces of the dye observed. For a quantitative check, the absorbance of the sample could have been measured using a spectrophotometer [26]. An absorbance closer to zero would have been achieved after clear liquid was dispensed.

Pipette

- Target volumes ranging for 100 μ L and 1 mL are set

- Liquid is dispensed and output is measured using a scale
- The actual volume against the target volume is recorded
- Pipette is calibrated to minimize error
- Dyed liquids are run through the system
- Pipette tip is removed and replaced
- Cycle is repeated and run with clear liquid to confirm no carryover from previous cycle

The system was programmed to dispense volumes between 100 μ L and 1 mL and was verified to be within a tolerance of $\pm 1\%$. Tests were conducted with various inorganic chemicals and dispensing speed was assessed by observing liquid's flow rate and splash.

Gantry System

- Measure positional accuracy by comparing actual position to target position
- Record deviation from target positions for each axis
- Test for repeatability by programming gantry to return to a target point over multiple cycles
- Test for the speed of movement by using the micros() Arduino function and measuring the distance

The accuracy, precision and repeatability of movements with the gantry were assessed across multiple cycles. Tests were conducted under standard lab conditions (25°C, 298K, 1 atm) with minimal positional error.

Display

- Confirm the default screen is displayed when connected to Arduino
- Load a sample GUI via a microSD card and check the interface displays correctly
- Test accuracy of all GUI elements for touchscreen accuracy by observing output responses
- Qualitatively inspect the display to ensure there is uniformity in the pixel light intensity
- Run the display for several hours to check for stability, overheating or glitches

4.1.3 User Testing and Feedback

Due to the limited time after completion of the project only a 20 people were able to participate in testing of the product. This also made it difficult to find many participants with physical disabilities. If there was more time after finalising the product, the group would've ensured a larger sample size, that was representative of the target market, was used.

To compensate for the lack of representation in the sample, finger resistance bands were used on the able-bodied participants to limit their mobility and dexterity in the hands. It was also confirmed by our disability consultant (an engineer with chronic pains, muscles spasms and hand tremors) that this was an adequate representation of what limited mobility can feel like for some disabilities.

Overall, the feedback from participants was positive (Appendix J), with 90% of participants rating the aesthetics of the design 4-5. This suggests that most people thought the product would fit seamlessly into a lab environment. However, there was a range of results for the portability of the design - 45% of participants thought the machine would be easy to move

under less intense periods of limited mobility. 100% of participants found no difference in difficulty between dispensing and aspirating the liquid. Further comments repeatedly suggested either a larger touchscreen or bigger touch icons were used as they found it slightly more difficult to navigate the user interface. Some disabled participants also suggested that tactile feedback from buttons or a keypad would have been preferred.

4.1.4 Future improvements to design

With more project time and resources, we would implement the following:

Improvement	Outcome
Add a mechanical or servo-driven pipette tip ejection system that presses the side button of the pipette	Our current prototype lacks a tip ejection mechanism, requiring manual removal. Incorporating a linear actuator or servo to press the side button would enable hands-free ejection. This significantly improves accessibility for users with limited dexterity, ensures biosafety by reducing contamination risk, and aligns with good laboratory practice (GLP) standards.
Design a compact, integrated tip disposal unit (e.g. mini sharps bin)	A small, enclosed container positioned adjacent to the pipette holder would allow automatic ejection of tips directly into a safe place. This promotes hygienic disposal, prevents biohazard exposure, and complies with lab waste management protocols. The bin should be removable for easy emptying and should be autoclavable or disinfectable.
Replace 3D-printed structural components with CNC-machined or injection-moulded parts for final-stage manufacturing	Transitioning to more robust manufacturing methods in large-scale production improves structural integrity, thermal and chemical resistance, and aesthetic finish. Injection moulding offers lower per-unit cost at scale, while CNC machining provides tighter tolerances and durability for moving parts.
Include a removable spill tray or hydrophobic barrier beneath the pipette tip	This protects critical electronics from liquid spills or pipette drips. A removable tray facilitates cleaning, while a hydrophobic coating or physical splash guard further reduces the risk of short-circuiting, enhancing device longevity and electrical safety.
Develop a fully enclosed, modular outer casing	Enclosing wires and electronics in a custom-fitted case improves user safety, reduces the chance of accidental disconnections or electrostatic discharge, and gives the system a professional, medical-grade appearance. Modular design allows future subsystems (e.g., automated tip racks, wireless controls) to be added without redesigning the full enclosure.
Implement noise- and vibration-reduction strategies	Our Motor and gantry systems currently generate significant noise. Damping techniques would reduce mechanical vibration and acoustic interference, enhancing comfort in quiet labs and increasing pipetting precision.
Upgrade UI system: larger display, tactile feedback, and alternative input methods	Replace the current small touchscreen with a larger capacitive display to improve visibility and touch accuracy. Redesign the UI to bigger icons, clearer layouts, and colour-contrast options. Incorporate tactile and/or auditory feedback to support users with visual impairments or sensory sensitivities.
Integrate a high-throughput batch pipetting mode	Currently, each pipetting action must be triggered individually. By introducing programmable sequences throughput can be significantly increased. This would be useful in repetitive workflows.

Table 2. Table of Future Improvements

4.2 Project Reflection

4.2.1 Core Values

From the beginning of the project, we wanted to create a safe and welcoming environment for all members of the group to respectfully and unapologetically share their opinions of our work. This made us all very comfortable with having our voices heard and also receptive to constructive criticisms. When members of our group had conflicting ideas, we were encouraged to critique each other and debate on what the next steps in our project should be. In addition to this, democratic votes were regularly taken via polls to ensure the majority of members in the group were happy with decisions being made. Even when polls were taken, we were all still comfortable further debating our ideas to make sure the best possible decisions were made for the future of our project. This was extremely beneficial to our as our consistent collaborative efforts created even better ideas than what we initially had.

It was imperative that each member in our group was clearly communicated with and understood the different stages of our project. During our initial design ideas phase, we would all draw rough sketches of how we interpreted and idea from another member of the group. This allowed us to clarify the ideas of our members and ensured we had to correct impression of what they were saying. By doing this, we were able to make well-informed decisions on the future of our project. When we all drew sketches of design ideas, it allowed for more effective communication of member's individual perspectives which we would later bring together to make a more interesting product. This created a lot more cognitive diversity in our group which is essential for any successful group project.

4.2.2 Division of the Workload

We all initially discussed our strengths and weaknesses with regards to the skills required for our group project. We believed that it was important to align tasks with team members' competencies to maintain high standards of work. The aim of our project is to increase accessibility for disabled people in the lab, meaning that it was important for us to reach out to as many people as possible from these communities to get their perspective on our designs. Because of this, all members of our group were initially assigned outreach and field research tasks to gather as much information as possible before the design work commenced. After this, we were divided into subgroups based on our interests, the demand of specific design components and the urgency for when things needed to be completed. For example, initially more team members worked on the PSD due to its urgency - this later transitioned to more team members working on the mechanics and electronics of the gantry system as this was a main component of our design. The subgroups that were initially set were fluid and not at all homogenous. We made sure that when work had been completed, team members moved to other areas of the projected as and when they were needed.

4.2.3 Means of Communication

Our WhatsApp group and Teams were the main platforms used to maintain regular communication with other team members. We would post video updates to document the progression of different components of the design so that all members of the group were up to date with the status of the project. There were also sub-WhatsApp groups where more specific conversations were had, pertaining to the different components of this design. This allowed us to easily compartmentalize the information and progression related to different parts of the project to avoid confusion. We would also have meetings throughout the year via WhatsApp, Microsoft Teams, or in person. The minutes of the meetings were recorded

on OneNote so we could easily refer back to what was discussed. As well as this, we also used Google Docs, Word and Canva to easily integrate individual's work.

4.2.4 Improvements

There was a very amicable and encouraging team dynamic that only bolstered very productive work for the duration of the project. However, there are always points of improvement that need to be considered in the future projects we all undertake. Although we communicated quite frequently on other platforms, there should have been a more regular schedule for weekly or biweekly in-person meetings to discuss the status of the group work. More in-person meetings may have led to more effective communication and allowed for clarification in case anything was misinterpreted online. For the meeting that we did have, recordings should have been in case the minutes on OneNote were unclear or lacked detail. To better organize meetings in future projects we aim to incorporate other platforms that better facilitate scheduling, such as Slack or Monday.

4.3 Conclusion

In conclusion, the automated micro pipetting system successfully achieved its core aim: enabling individuals with limited dexterity to independently and accurately perform pipetting tasks, thereby enhancing accessibility in laboratory environments. While some features remain in development or require refinement, such as tip ejection and expanded user feedback, the overall system met key design and performance criteria, including accuracy, usability, and affordability. Through iterative testing, user input, and thoughtful design, the project demonstrated a viable and scalable assistive technology solution. This work not only contributes a practical tool for inclusive education and research but also highlights the broader value of human-centered design in promoting equity within STEM fields.

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Appendices

Appendix A – Project Management

Our team consisted of 9 2nd year Molecular Bioengineering students. and was supervised by Dr Ian Radcliffe. Responsibilities were distributed based on individual group members' personal interests and skills. The team was initially divided into sub-groups to manage different components of the design: pipette, electronics, gantry mechanics and CAD. Each sub-group provided regular updates during meetings held via Microsoft Teams, with meeting minutes recorded on a shared OneNote notebook and communication was maintained through WhatsApp.

As a collective, the group generated initial design ideas that were eventually refined until a final concept was agreed upon. As apart of this process, all members took part in initial research and data collection – liaising with disability officers and lab technicians to get their input on our ideas so informed decision could be made on the direction of the project.

The distribution of roles across the group are as follows:

All group members contributed to the PSD, presentation, poster and prototyping.

Ben Jones

- PSD Editor
- CAD Technician
- Final Report Writer

Sheena Ling – Procurement Manager

- Poster writer and designer
- Final design drawings
- CAD Technician

Jenna du Preez – Scribe

- PSD Editor
- 3D printing
- Poster content

Sumaiyah Zaman

- PSD Editor
- Report Writer
- 3D printing

Kailin Wang

- Gantry Assembler

Tayeb Tadlaoui

- Presentation Video Editor
- LCD Touchscreen Developer
- Arduino code Developer

Leonardo Grimaldi

- Gantry Manufacturer
- Arduino code Developer

Yasmine Marenzi – Project Manager

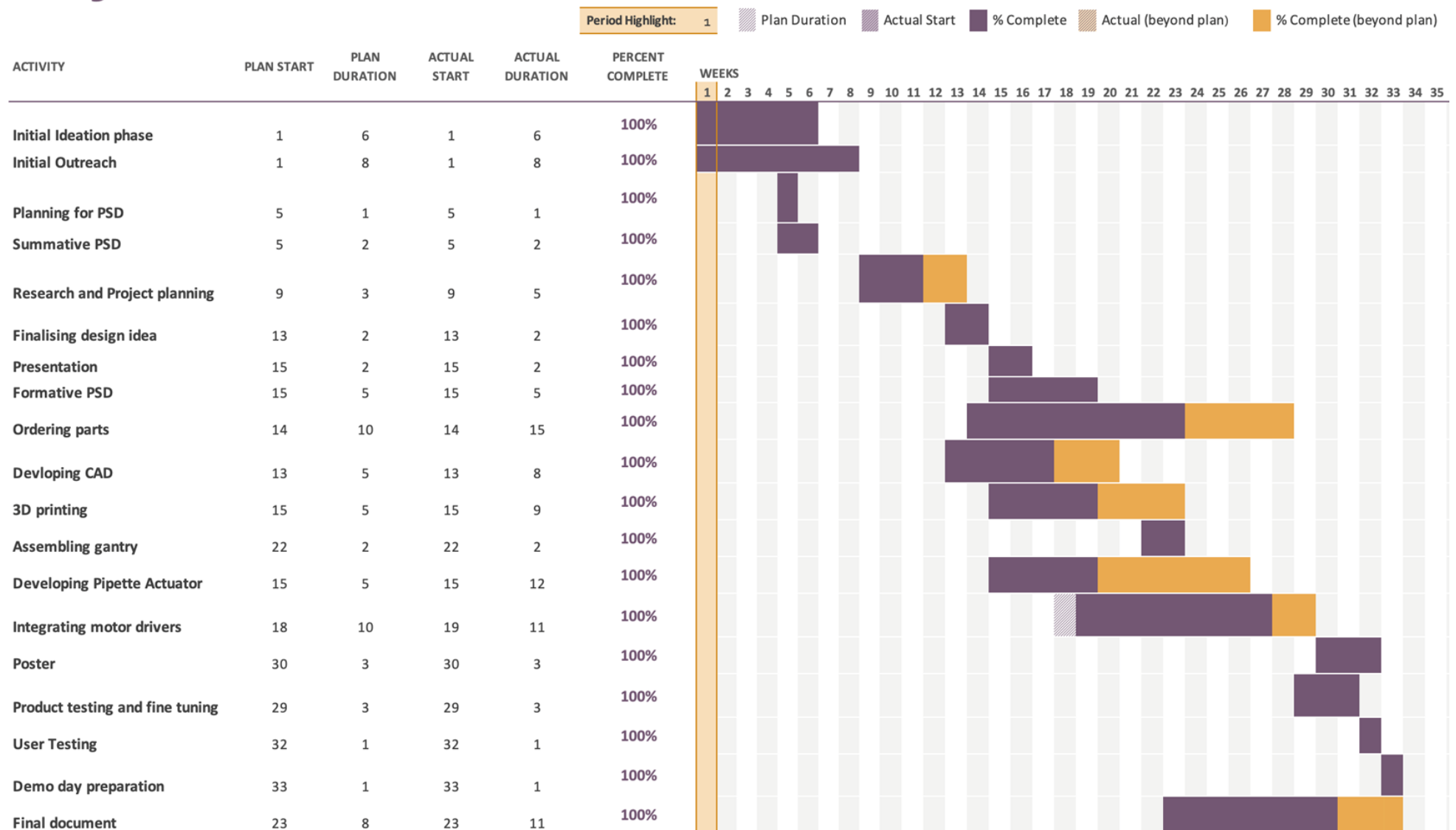
- Pipette and Actuator Manufacturer
- CAD and 3D printing
- Arduino Code Developer

Sean Obi

- Final Report Writer
- Electronics Leader
- Arduino code Developer
- Gantry Assembler
- User-testing coordinator

Figure A.1: Gant Chart

Project Planner



Appendix B – Risk Management

Potential risks or causes of failure in our design were analysed and rated based on their Severity (S), probability of Occurrence (O) and Detection (D). Each of these 3 components to the risk were evaluated on a scale of 1-10. The 3 numbers are multiplied to achieve the Risk Priority Number (RPN) to define the level of risk posed by particular parts of the design. A critical RPN threshold of 75 was chosen and preventing measure were put in place to ensure the RPN stays well below the threshold.

S – Severity

Rating S	Criteria: Severity of effect	Consequence	Treatment
10	Death	-	-
9	Quadriplegia	Life-long medical care necessary / coma / permanent damage	Hospital stay
8	Amputations, paraplegia, blindness, deafness, traumatic brain injury (severe), fourth-degree burns	Life-long medical care necessary / coma / permanent damage	Hospital stay
7	Complex fractures, open fracture, inner injuries, traumatic brain injury (severe), third-degree burns	Permanent damage possible	Hospital stay
6	Gash, fractures, torn muscles, articular cartilage injury, traumatic brain injury (moderate), second-degree burns	Permanent damage possible	Hospital stay
5	Gash, fractures, torn muscles, articular cartilage injury, traumatic brain injury (mild), second-degree burns	Reversible injury	Hospital stay or ambulant treatment
4	Severe cuts, severe scratches, severe contusions, strains, first-degree burns	Reversible injury	Ambulant treatment or self-treatment
3	Minor cuts, minor scratches, minor contusions, stiff muscles, tension, blisters, excoriations, sickness, first-degree burns	Discomfort during application up to three days after application	Self-treatment
2	Slight sickness, pressure marks	Discomfort	-
1	No harm	-	-

O – Occurrence

Rating O	Criteria: Probability of occurrence
10	Occurs or may occur very likely during every use of the session
9	Occurs or may occur likely during every use of the session
8	Occurs in 1 of 5 sessions (less than once a day)
7	Occurs in 1 of 10 sessions (less than once a day)
6	Occurs in 1 of 50 sessions (less than once half a month)
5	Occurs in 1 of 100 sessions (less than once a month)
4	Occurs in 1 of 500 sessions (less than once half a year)
3	Occurs in 1 of 1000 sessions (less than once per year)
2	Occurrence very unlikely
1	Occurrence nearly impossible

D – Detection

Rating D	Criteria: Likelihood of detection by design control
10	No chance of detection
9	Very remote chance of detection
8	Remote chance of detection
7	Very low chance of detection by indirect methods (hardware or software)
6	Low chance of detection by indirect methods (hardware or software)
5	Moderate chance of detection by indirect methods (hardware or software)
4	High chance of detection by indirect methods (hardware or software)
3	High chance of detection by direct or indirect methods (hardware/software)
2	Direct and indirect detection: Hardware or software
1	Direct detection: Hardware or safe software (category 4, performance level e)

Risk Analysis

Assembly	Failure & Effect	S1	O1	D1	RPN before	Preventing measures	S2	O2	D2	RPN after
Pipetting Mechanism	The pipette actuator is incorrectly calibrated and leads to incorrect volume dispensed. Risks experimental error and could lead to liquid spillage, damaging electronics.	2	10	8	160	Pipetting mechanism should be calibrated before integration into system. Software-driven calibration can also be implemented for the actuator. A test dispense feature should be implemented before a new aliquoting cycle.	1	2	2	4
Touchscreen Interface	Graphical distortions (flickering/flashing) on the screen may trigger disabilities (epilepsy) and confuse the user with regards to functionality.	5	5	3	75	Design a user interface with stable refresh rate and bigger icons. Avoid any flashing animations and consult the accessibility standards in WCAG during the UI design.	1	2	3	6
Touchscreen Lag	It is possible to misinterpret the touch icons when the response is delayed. This increases the time taken to carry out experiments and can increase the fatigue or stress put on the user.	4	7	5	140	Add a keypad of buttons that allow users to operate the device without the need for the touchscreen. This can be used as a failsafe.	2	7	5	70
Gantry Collision	Moving gantry system may collide with the frame or nearby objects (users' hands). This can potentially damage different mechanical components or cause user injury.	8	3	6	144	Limit switches are added to act as physical and logical boundaries in firmware that prevent the overextension of the gantry. The end-stop sensors in the limit switches will halt the movement of the gantry and trigger a return-to-origin sequence.	2	2	2	8
Pipette tip handling	Caused by residual fluid or reuse of pipette tips, leading to cross contamination of samples which can compromise the integrity of results and the sample purity.	1	8	9	72	Mandate single-use pipette tips and implement training for the user. Add visual prompts in the UI for manual tip replacements. In future developments, add automatic tip ejection.	1	2	9	18

Tip attachment mechanism	Pipette tip fails to attach securely, causing tip to fall off during aspiration or dispensing, contaminating samples and reducing experimental reliability.	6	6	4	144	Implement a mechanical locking system to secure the tip and verify attachment with a load sensor or pressure check in software. Instruct users to visually inspect tip placement before use.	3	2	2	12
Long term actuator wear	After repeated cycles, the linear actuator may wear out, reducing precision or causing erratic movements, risking sample loss or injury.	4	4	5	80	Use an actuator rated for high cycle life. Include periodic maintenance guidelines. Optionally allow users to replace actuator easily via a modular design.	2	2	5	20
Internal electronics overheating (Fire hazard)	Prolonged use of the device can lead to overheating of the Arduino and the motor drivers. When components exceed safe operating temperatures this can lead to permanent damage of components which inhibits the functionality and consequently reduces the lifespan of the product. If ignition temperatures are reached this can cause melting, smoking or even fires.	9	3	5	135	Ensure experiments are conducted in a well-ventilated room for better air circulation. Add heat sinks to passively cool the electrical components by releasing the excess heat to the surroundings. Regularly use freeze spray for a rapid, local decrease in temperature on the circuit board.	1	3	5	15
Overvoltage (Mains)	If mains power (230V AC) is improperly connected or regulated it can exceed the safe operating voltage for low-voltage DC components, causing instant failure or even a fire risk in the device.	9	2	7	126	Use a certified external regulated adapted (e.g. 12 V) and step down the voltage to 5 V using buck converters. Add a fuse to break the circuit when current gets too high. Instead, use bench top power supplies for each motor driver – allows for precise voltage and current control.				
Overvoltage (Bench top power supply)	Incorrect voltage or current setting on the external power supply may result in overvoltage which could damages the electrical components or pose a	5	8	1	40	Train users to verify voltage and current settings before use. Cover unused outputs with insulating caps. Add a fuse to break the circuit if current gets to high. Label all	5	4	1	20

	short-circuit risk, especially in enclosed/cluttered lab spaces.					output ports with the allowed voltage and current (10 V, 300 mA)				
Electronic Housing	Absence of proper casing or enclosure for the electronics may allow liquids from pipetting to spill onto electrical components (Arduino, wires and motor drivers) and cause short circuits.	2	3	8	48	Add a sealed acrylic cover over all exposed electronics. Move circuit boards away from fluid handling zones. Add splash guards to walls of the gantry system.	2	2	8	32

Table B.1: Table of Risk assessments

Appendix C – Ethics

- **Ethical considerations**

Ethical considerations were an integral part of the design and development of our Automated Micro pipetting System, given the device's intended purpose of improving accessibility for individuals with physical disabilities. Our aim was not only to create a technically functional device, but also to ensure that it aligns with broader principles of inclusivity, respect for users, and responsible engineering practice.

- **Honesty**

Throughout our project, we ensured that all data, research, surveys, and technical limitations were communicated truthfully, especially when reporting the system's accuracy and limitations in our final testing. (Add an example after user testing done)

- **Scientific Integrity**

All literature used during the development of our kit was thoroughly cited and acknowledged in our final report. We upheld academic integrity by giving full credit to all open-source designs incorporated into our project, including the 3D-printed micropipette model and the Ender 3 gantry system. Any ideas, data, or designs that informed or inspired our work were appropriately referenced to ensure transparency and respect for intellectual property.

- **Legality**

We ensured the design complied with standard safety guidelines, any images or videos used did not violate copyright or licensing laws and used only legal sources for components and code. To be released onto the market, our Automated Micro pipetting System would need to be registered with the MHRA in compliance with UK Medical Devices Regulations. ([Register medical devices to place on the market - GOV.UK](#))

- **Respecting Intellectual Property**

We ensured any sample code, CAD models, or mechanical parts adhered to licensing agreements. For example, the open-source micropipette we adopted complies with ISO standards and its licensing terms.

- **Respect for Colleagues**

To ensure fairness and inclusivity within the team, tasks were divided as equally as possible. Open communication was encouraged through regular meetings, polls, and discussions, creating a respectful environment where all opinions could be shared and constructive feedback welcomed. Group decisions were made collaboratively, with polls conducted via WhatsApp and Teams, and majority votes guiding outcomes. Personal commitments, such as health needs, were accommodated, and members supported one another during challenging periods.

- **Non-Discrimination**

Through research and expert consultation, we identified hand tremors and dexterity impairments as key barriers to equal participation in manual micro pipetting. Our design

prioritized inclusivity by creating a system accessible to users with varied physical abilities, including limited dexterity, tremors, or reduced hand strength. We followed established accessibility guidelines and used inclusive language to ensure no user is excluded or disadvantaged based on physical ability. This reflects our ethical commitment to non-discrimination, removing barriers to fair access, and enabling all individuals to participate fully in laboratory work.

- **Privacy and Confidentiality**

User testing, feedback and survey responses were fully anonymized to protect participants' identities. No personal data was collected or stored, and all feedback was treated confidentially. Informed consent was obtained where applicable, ensuring ethical handling of all user interactions.

- **Environment & Sustainability**

Environmental impact and sustainability were core considerations throughout the design and development of our Automated Micro pipetting System. Our goal was to create an accessible, functional device while actively minimizing energy consumption, material waste, and the system's overall environmental footprint.

Various structural components, including the pipette mount and adaptors, were 3D printed using PLA, a biodegradable material under industrial composting conditions. We optimized print settings such as infill percentage and layer height to reduce filament use without compromising durability. Multiple parts were printed simultaneously in batch jobs to reduce energy consumption from the printer. Failed prints were disposed of responsibly and recycled where possible.

Electronic components such as the Elegoo Mega 2560, motor drivers, and actuators were sourced from UK-based or domestic suppliers when available to reduce carbon emissions associated with long-distance shipping. Orders were combined to reduce packaging waste and delivery frequency.

To reduce future waste, the system was designed with modularity and repairability in mind. Components such as the pipette tip holder, linear actuators, and end stop switches can be individually replaced or upgraded without discarding the full unit. This extends the product's lifespan and supports a circular usage model. Where feasible, we selected reusable or recyclable materials and adhered to WEEE guidelines for responsible e-waste management.

The device also considered sustainability during use. Its lightweight frame and efficient actuator system limit unnecessary energy consumption. Used micropipette tips are intended to be disposed of according to standard lab safety and environmental waste protocols, including sharps bin use where relevant.

Appendix D - Bill of Materials

Component	Manufacturer Part Number (MPN)	Description	Manufacturer	Suppliers	Units	Unit Cost /£	Total Cost	Remarks
Electronics (Arduino, PCB)	EL-CB-003	Elegoo Mega 2560 R3 Controller Board	Elegoo	Amazon	1	21.98	21.98	Arduino Mega Compatible
	N/A	2-Layer Bare Rigid Printed Circuit Board	JLCPCB	JLCPCB	1	99.22	99.22	Customised PCB board
	PPTC081LFB N-RC	CONN HDR 8POS 0,.1 TIN PCB	DigiKey	DigiKey	19	0.496	9.42	/
	277-1258-ND	Term Blk 2Pos Side Entry 5mm PCB	DigiKey	DigiKey	1	1.2	1.20	2-Position Side-Entry Terminal Block, 5.08mm Pitch
	552440	LM2596 DC-DC Switching Adjustable Step Down Voltage Regulator Module 4.5 V-40V	Switch Electronics	Switch Electronics	5	1.95	9.75	/
	D-A4988	Control Stepper Motor with A4988 Driver Module and Arduino	Oosznest	Oosznest	4	4.52	18.08	Max output of 35 V at +/- 1.5A, with 5 step revolutions of full, half, quarter step, etc. Built in potentiometer.
Gantry System	Ender-3	Creality Ender-3 3D Printer	Creality	Creality Official Store	1	139.00	139.00	Includes frame, motion system, power supply, control system, LCD interface and wiring
Pipette	732489269615	10 Pack 3ml/3cc Syringe	GSSFASHION	Amazon	1	5.48	5.48	Made of PLA, PC and PE. Dimensions: 8.7L x 2.5W cm.
	602451234203	Sourcingmap M3 x 35mm Carbon Steel	RC Screws	Amazon	1	6.59	6.59	Grade Rating: Class 10.9

		Full Thread Hex Socket Cap Head Screws Bolt 25 Pcs						
	B08HL9VVHX	M3 Hex Full Nut - A2 Stainless Steel (20 pcs)	Ascot City	Amazon	1	6.49	6.49	DIN 934
	LC022D10M	Standard Compression Spring	Lee Spring	Lee Spring	2	4.24	8.48	/
	E-F-160049	Kynar Luer, Female Luer Lock to 5/32" (4.0mm) ID Tube	Eoxshop	Eoxshop	1	27.36	27.36	Meets Requirements for USP Class VI, California Prop. 65, ISO 594
Screen	296683638009	Nextion 4.3" HMI Display Basic NX4827T 043 LCD-TFT Resistive Touch Screen 480x272	SONOFF STORE	Ebay	1	52.79	52.79	Compatible with NEXTION editing software
	B0DRNNH8JR	32GB Micro SD Card, Memory Card with SD Adapter	PHISON ELECTRONI CS CORP	Amazon	1	11.98	11.98	/
Actuator	RB-Fir-121	Actuonix Linear Actuator Control Board	RobotShop	RobotSh op	1	10.01	10.01	/
	2595	Kitronik Linear Acuator Micro Servo Kit	Kitronik	Kitronik	1	18.95	18.95	/
Maintenance/Testing	741416340358	4 Pack Hand Exerciser Grip Strengtheners for Arthritis	PORTHOLI C	Amazon	1	13.98	13.98	/
	M090409	Motip Freezer Electronic Spray - 360ml	MOTIP	Amazon	1	15.28	15.28	Non- flammable
	5010305061236	Loctite Universal Strong All Purpose Super Glue	Loctite	Amazon	1	7.49	7.49	/
Total Cost (£)		483.53						

Table D.1: Bill of material

Appendix E – Nomenclature

Actuator – A component that moves or controls a mechanism or system (e.g., applies force).

Arduino – An open-source electronics platform based on easy-to-use hardware and software, used in embedded systems development.

Buzzer – An electronic component that emits a sound signal, typically used for auditory feedback.

CAD – Computer-Aided Design.

CNC – Computer Numerical Control, a manufacturing process that uses computerized controls to operate machine tools.

Chemical Compatibility – The ability of a material or system to resist chemical degradation when exposed to various substances.

Display (Touchscreen / Touchpad) – A display that also serves as an input device by responding to touch.

FDM – Fused Deposition Modelling, a 3D printing technique.

GLP – Good Laboratory Practice, a set of principles intended to ensure the quality and integrity of non-clinical laboratory studies.

Gantry System – A mechanical structure that allows controlled movement along multiple axes (X, Y, Z).

GUI – Graphical User Interface.

I/O – Input / Output, referring to the transmission of data.

IDE – Integrated Development Environment.

IEC – International Electrotechnical Commission.

ISO – International Organization for Standardization.

LCD – Liquid Crystal Display.

Labware – Reusable or disposable items used in laboratories (e.g., beakers, tubes, pipette tips).

Limit Switch – A sensor that signals when a component has reached the end of its range of movement.

LAC – Linear Actuator Control.

mL (Millilitre) – One-thousandth of a litre (10^{-3} litres), standard unit of liquid volume in laboratory contexts.

Microlitre (μL) – One-millionth of a litre (10^{-6} litres), used for measuring small liquid volumes in labs.

Micros() – An Arduino function that returns the number of microseconds since the board began running the current program.

PCB – Printed Circuit Board.

PLA – Polylactic Acid, a biodegradable thermoplastic used in 3D printing.

PWM – Pulse Width Modulation.

Spectrophotometer – An analytical instrument used to measure the amount of light absorbed by a sample, often for quantitative chemical analysis.

Stepper Motor – A type of motor that moves in discrete steps, often used for precise control of position.

Systematic Error – Consistent, repeatable error associated with faulty equipment or flawed technique.

UART – Universal Asynchronous Receiver/Transmitter, responsible for serial communication within an embedded system.

WCAG – Web Content Accessibility Guidelines.

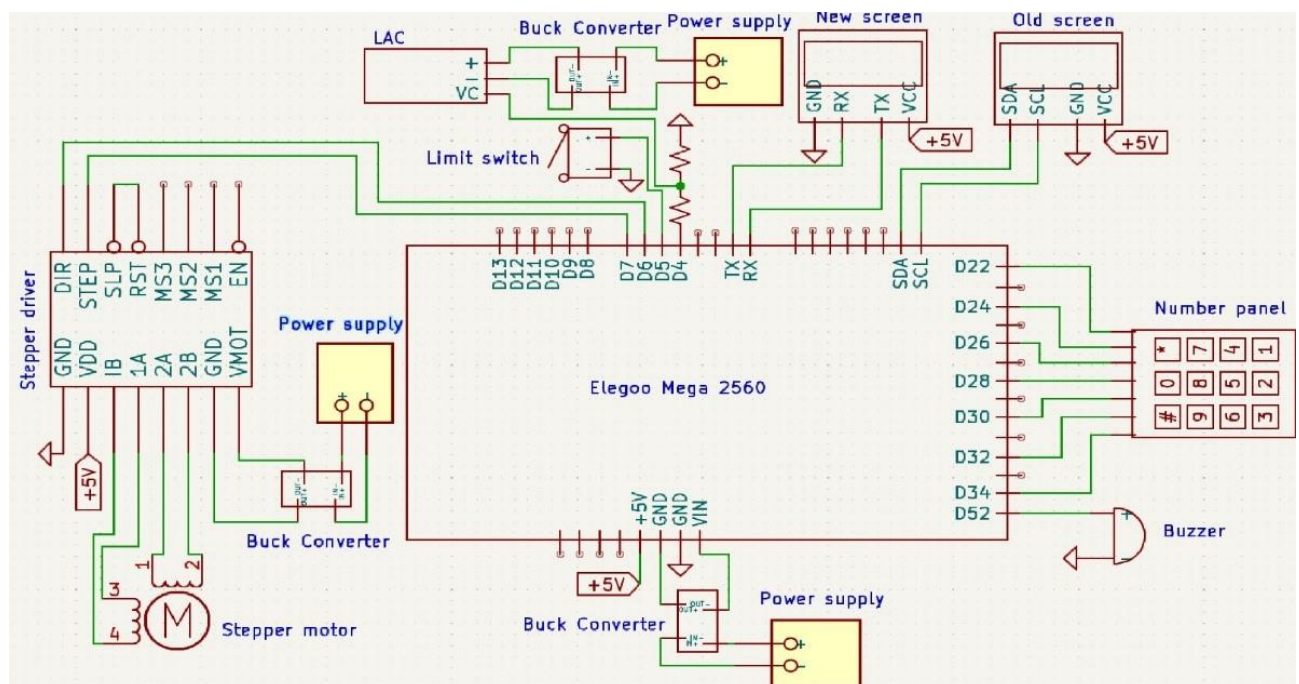
Appendix F – Electrical Schematics

Appendix F presents the complete electrical schematic for the Automated Micropipetting System. The diagram illustrates the connections and integration of all the key components within the system including: Elegoo Mega 2560 microcontroller, stepper motor drivers, LAC board, power regulation units and user interface components (touchscreen display and numerical keypad), limit switches and auxiliary devices such as the buzzer.

The schematic is organised to show the data transmission and power connections:

- Power supply management: dedicated 12 V (LAC board) and 5 V (Stepper motor drivers, microcontroller, screen) regulated rails to ensure stable operation of components.
- Motor control: Stepper interfaced with microcontroller for co-ordination of movement of gantry system
- Actuator Control: PWM controlled linear actuator managed via LAC board
- User interface: UART-connected touchscreen display for primary user feedback, I2C auxiliary display (no longer used) and a numerical keypad for user inputs.
- Safety Features: End stop switches present on each axis for homing and positional reset protocols, with direct inputs to microcontroller for real time adjustments

Figure F.1: Complete electrical schematic of Automated Micropipetting System, detailing microcontroller connections, power management, and user I/O connections



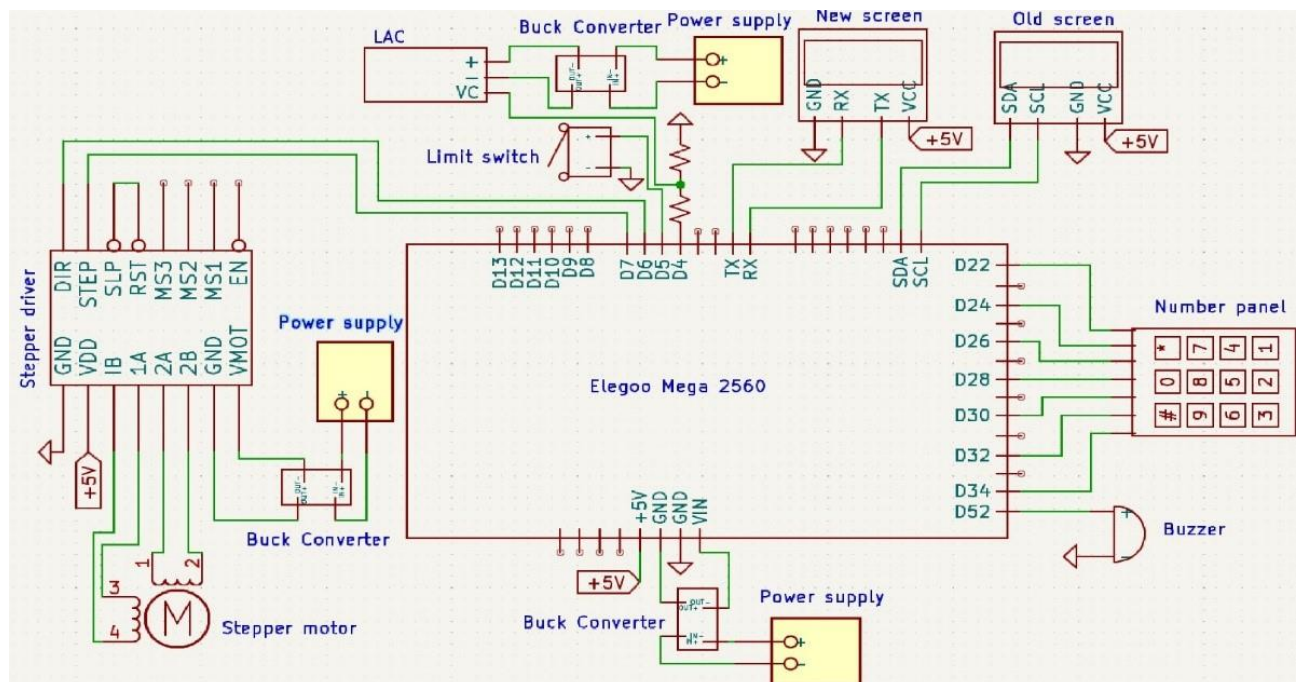
Note: visualised using KiCad.

complete electrical schematic for the Automated Micro pipetting System. The diagram illustrates the connections and integration of all the key components within the system including: Elegoo Mega 2560 microcontroller, stepper motor drivers, LAC board, power regulation units and user interface components (touchscreen display and numerical keypad), limit switches and auxiliary devices such as the buzzer.

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Figure F.1: Complete electrical schematic of Automated Micropipetting System, detailing microcontroller connections, power management, and user I/O connections



Note: visualised using KiCad.

Appendix G – Code

Figure G.1: Arduino Code

```
1  #include <Arduino.h>
2
3  // --- Pin Definitions ---
4  constexpr uint8_t ENDSTOP_PIN_X = 5;
5  constexpr uint8_t DIR_PIN_X     = 7;
6  constexpr uint8_t STEP_PIN_X    = 6;
7
8  constexpr uint8_t ENDSTOP_PIN_Y = 8;
9  constexpr uint8_t DIR_PIN_Y     = 10;
10 constexpr uint8_t STEP_PIN_Y    = 9;
11
12 constexpr uint8_t ENDSTOP_PIN_Z = 11;
13 constexpr uint8_t DIR_PIN_Z     = 13;
14 constexpr uint8_t STEP_PIN_Z    = 12;
15
16 constexpr uint8_t BUZZER_PIN = 2; //  Buzzer pin added
17
18 constexpr int LIMIT_PRESSED = HIGH;
19 const long MAX_BACKWARD_XY = 1100;
20 const long MAX_BACKWARD_Z = 1700;
21 const long MAX_HOMING_STEPS_Z = MAX_BACKWARD_Z + 100;
22 const long Z_SAFE_HEIGHT = 1700;
23 const long Z_WORK_HEIGHT = 300;
24 unsigned long stepDelayXY = 1500;
25 unsigned long stepDelayZ = 1200;
26
27 long stepPositionX = 0;
28 long stepPositionY = 0;
29 long stepPositionZ = 0;
30
31 constexpr uint8_t DIR_TOWARD_Z = HIGH;
32 constexpr uint8_t DIR_AWAY_Z = LOW;
33
34 struct Coord { long x; long y; };
35
36 Coord wells[] = {
37   { 890, 500 }, { 940, 500 }, { 990, 500 }, { 1040, 500 }, { 1090, 500 },
38   { 890, 550 }, { 940, 550 }, { 990, 550 }, { 1040, 550 }, { 1090, 550 },
39   { 890, 600 }, { 940, 600 }, { 990, 600 }, { 1040, 600 }, { 1090, 600 },
40   { 890, 650 }, { 940, 650 }, { 990, 650 }, { 1040, 650 }, { 1090, 650 },
41   { 890, 700 }, { 940, 700 }, { 990, 700 }, { 1040, 700 }, { 1090, 700 },
42   { 890, 750 }, { 940, 750 }, { 990, 750 }, { 1040, 750 }, { 1090, 750 },
43   { 890, 800 }, { 940, 800 }, { 990, 800 }, { 1040, 800 }, { 1090, 800 },
44   { 890, 850 }, { 940, 850 }, { 990, 850 }, { 1040, 850 }, { 1090, 850 }
45 };
46
47 Coord beakers[] = {
48   { 400, 700 },
49   { 670, 700 }
50 };
51
52 #define nextion Serial1
53 const size_t BUF_SIZE = 128;
54 uint8_t buf[BUF_SIZE];
55 size_t idx = 0;
56
57 void pulseStepX() {
58   digitalWrite(STEP_PIN_X, HIGH); delayMicroseconds(stepDelayXY);
59   digitalWrite(STEP_PIN_X, LOW);  delayMicroseconds(stepDelayXY);
60 }
61
62 void pulseStepY() {
63   digitalWrite(STEP_PIN_Y, HIGH); delayMicroseconds(stepDelayXY);
64   digitalWrite(STEP_PIN_Y, LOW);  delayMicroseconds(stepDelayXY);
65 }
66
67 void pulseStepZ() {
68   digitalWrite(STEP_PIN_Z, HIGH); delayMicroseconds(stepDelayZ);
69   digitalWrite(STEP_PIN_Z, LOW);  delayMicroseconds(stepDelayZ);
70 }
71
72 void moveToXY(long targetX, long targetY) {
73   moveToZ(Z_SAFE_HEIGHT);
74
75   long dx = targetX - stepPositionX;
76   long dy = targetY - stepPositionY;
77
78   int dirX = dx >= 0 ? 1 : -1;
79   int dirY = dy >= 0 ? 1 : -1;
80
81   digitalWrite(DIR_PIN_X, dirX > 0 ? HIGH : LOW);
82   digitalWrite(DIR_PIN_Y, dirY > 0 ? HIGH : LOW);
```

```

81
82 dx = abs(dx);
83 dy = abs(dy);
84
85 long steps = max(dx, dy);
86 long errorX = 0, errorY = 0;
87
88 for (long i = 0; i < steps; ++i) {
89     errorX += dx;
90     errorY += dy;
91
92     if (errorX >= steps) {
93         pulseStepX();
94         stepPositionX += dirX;
95         errorX -= steps;
96     }
97
98     if (errorY >= steps) {
99         pulseStepY();
100         stepPositionY += dirY;
101         errorY -= steps;
102     }
103
104     delayMicroseconds(stepDelayXY * 2);
105 }
106
107 void moveToZ(long targetZ) {
108     if (targetZ == stepPositionZ && digitalRead(ENDSTOP_PIN_Z) != LIMIT_PRESSED) return;
109     digitalWrite(DIR_PIN_Z, targetZ < stepPositionZ ? DIR_TOWARD_Z : DIR_AWAY_Z);
110     long steps = abs(targetZ - stepPositionZ);
111     for (long i = 0; i < steps; ++i) {
112         if (digitalRead(ENDSTOP_PIN_Z) == LIMIT_PRESSED && targetZ < stepPositionZ) {
113             stepPositionZ = 0;
114             return;
115         }
116         pulseStepZ();
117         stepPositionZ += targetZ < stepPositionZ ? -1 : 1;
118     }
119 }
120
121 void holdZatWorkHeight() {
122     moveToZ(Z_WORK_HEIGHT);
123     delay(3000);
124     moveToZ(Z_SAFE_HEIGHT);
125 }
126
127 void homeAxisX() {
128     digitalWrite(DIR_PIN_X, LOW);
129     while (digitalRead(ENDSTOP_PIN_X) != LIMIT_PRESSED) pulseStepX();
130     stepPositionX = 0;
131 }
132
133 void homeAxisY() {
134     digitalWrite(DIR_PIN_Y, LOW);
135     while (digitalRead(ENDSTOP_PIN_Y) != LIMIT_PRESSED) pulseStepY();
136     stepPositionY = 0;
137 }
138
139 void homeAxisZ() {
140     digitalWrite(DIR_PIN_Z, DIR_TOWARD_Z);
141     for (int i = 0; i < MAX_HOMING_STEPS_Z && digitalRead(ENDSTOP_PIN_Z) != LIMIT_PRESSED; ++i) pulseStepZ();
142     stepPositionZ = 0;
143     moveToZ(Z_SAFE_HEIGHT);
144 }
145
146 // [X] Play confirmation beep (2 quick notes)
147 void playConfirmationBeep() {
148     tone(BUZZER_PIN, 1000, 100);
149     delay(150);
150     tone(BUZZER_PIN, 1500, 100);
151     delay(150);
152     noTone(BUZZER_PIN);
153 }
154
155 // [X] Play melody after homing
156 void playEndMelody() {
157     tone(BUZZER_PIN, 988, 150); delay(200); // B6
158     tone(BUZZER_PIN, 1319, 150); delay(200); // E7
159     tone(BUZZER_PIN, 1047, 200); delay(250); // C7
160     noTone(BUZZER_PIN);
161 }
162
163 void moveToWell(uint8_t n) {
164     if (n >= sizeof(wells)/sizeof(wells[0])) return;
165     moveToXY(wells[n].x, wells[n].y);
166     holdZatWorkHeight();
167 }
168
169 void moveToBeaker(uint8_t n) {
170     if (n < 1 || n > 2) return;
171     moveToXY(beakers[n-1].x, beakers[n-1].y);
172     holdZatWorkHeight();
173 }
174
175 void parsePacket(const uint8_t *data, size_t len) {
176     int inputWell = 0;
177     bool hasInputWell = false;
178     int outputWells[40];
179     int outCount = 0;
180     int volume = 0, beakerSide = 0, sourceType = 0;
181
182     memset(outputWells, 0, sizeof(outputWells));
183
184     // [X] Play confirmation tone
185     playConfirmationBeep();
186
187     for (size_t i = 0; i + 1 < len; i++) {
188         if (data[i] == 'I' && data[i+1] == '=') {
189             i += 2;
190             inputWell = 0;
191             hasInputWell = false;
192             while (i < len && isdigit(data[i])) {
193                 inputWell = inputWell * 10 + (data[i] - '0');
194                 hasInputWell = true;
195             }
196         } else if (data[i] == 'O' && data[i+1] == '=') {
197             i += 2;
198             while (i < len && data[i] != ';') {
199                 if (isdigit(data[i])) {
200                     int v = 0;
201                     while (i < len && isdigit(data[i])) v = v * 10 + (data[i] - '0');
202                     if (outCount < 40) outputWells[outCount++] = v;

```

```

201     } else i++;
202 }
203 } else if (data[i] == 'V' && data[i+1] == '=') {
204     if (i+6 <= len) volume = data[i+2] | (data[i+3]<<8) | (data[i+4]<<16) | (data[i+5]<<24);
205     i += 6;
206 } else if (data[i] == 'B' && data[i+1] == '=') {
207     if (i+6 <= len) beakerSide = data[i+2] | (data[i+3]<<8) | (data[i+4]<<16) | (data[i+5]<<24);
208     i += 6;
209 } else if (data[i] == 'T' && data[i+1] == '=') {
210     if (i+6 <= len) sourceType = data[i+2] | (data[i+3]<<8) | (data[i+4]<<16) | (data[i+5]<<24);
211     i += 6;
212 } else i++;
213 }
214
215 Serial.print("Input Well: "); Serial.println(inputWell);
216 Serial.print("Output Wells: "); for (int j = 0; j < outCount; ++j) { Serial.print(outputWells[j]); Serial.print(","); } Serial.println();
217 Serial.print("Volume: "); Serial.println(volume);
218 Serial.print("Beaker Side: "); Serial.println(beakerSide);
219 Serial.print("Source Type: "); Serial.println(sourceType);
220
221 if (sourceType == 1 && beakerSide >= 1 && beakerSide <= 2) {
222     moveToBeaker(beakerSide);
223     for (int i = 0; i < outCount; ++i) moveToWell(outputWells[i]);
224 } else if (sourceType == 2 && hasInputWell) {
225     moveToWell(inputWell);
226     for (int i = 0; i < outCount; ++i) moveToWell(outputWells[i]);
227 }
228
229 moveToZ(Z_SAFE_HEIGHT);
230 moveToXY(0, 0);
231
232 stepPositionX = 0;
233 stepPositionY = 0;
234
235 // 🟢 Play melody after return home
236 playEndMelody();
237
238 memset(buf, 0, sizeof(buf));
239 idx = 0;
240 }
241
242 void setup() {
243     Serial.begin(9600);
244     nextion.begin(9600);
245     delay(300);
246     while (nextion.available()) nextion.read();
247
248     pinMode(BUZZER_PIN, OUTPUT); // 🟢 Initialize buzzer pin
249
250     pinMode(ENDSTOP_PIN_X, INPUT_PULLUP); pinMode(DIR_PIN_X, OUTPUT); pinMode(STEP_PIN_X, OUTPUT);
251     pinMode(ENDSTOP_PIN_Y, INPUT_PULLUP); pinMode(DIR_PIN_Y, OUTPUT); pinMode(STEP_PIN_Y, OUTPUT);
252     pinMode(ENDSTOP_PIN_Z, INPUT_PULLUP); pinMode(DIR_PIN_Z, OUTPUT); pinMode(STEP_PIN_Z, OUTPUT);
253
254     homeAxisX(); homeAxisY(); homeAxisZ();
255     Serial.println("Ready to receive Nextion packets...");
256 }
257
258 void loop() {
259     while (nextion.available()) {
260         uint8_t b = nextion.read();
261         if (b == '#') {
262             if (idx > 0) {
263                 parsePacket(buf, idx);
264             }
265         } else if (idx < BUF_SIZE) {
266             buf[idx++] = b;
267         }
268     }
269
270     if (Serial.available()) {
271         String in = Serial.readStringUntil('\n');
272         in.trim();
273         if (in.equalsIgnoreCase("X0")) homeAxisX();
274         else if (in.equalsIgnoreCase("Y0")) homeAxisY();
275         else if (in.equalsIgnoreCase("Z0")) homeAxisZ();
276     }
277 }
278

```

Appendix H – User Instructions

1. Getting Started

1.1 Powering On

- Plug the system into a standard mains outlet (230V) to turn it on.
- Wait for the interface screen to initialize.

1.2 Preparing the Labware

- Ensure the source container (beaker or microcentrifuge tube) is filled with the desired liquid. This can be done by the user or a trained laboratory technician.
- Place your beaker or 1.5 mL microcentrifuge tubes in the clearly marked positions on the base of the device, similarly, done by the user or a trained technician.

2. Using the Touchscreen Interface

The interface guides the user step-by-step through the pipetting process. Below is an overview of each screen and its functionality. Use the Home button to return to the main screen at any time.

2.1 Page 0 – Source Type Selection

What you see:

- Image buttons/touch icons representing a beaker and tubes (Figure H1).

Action:

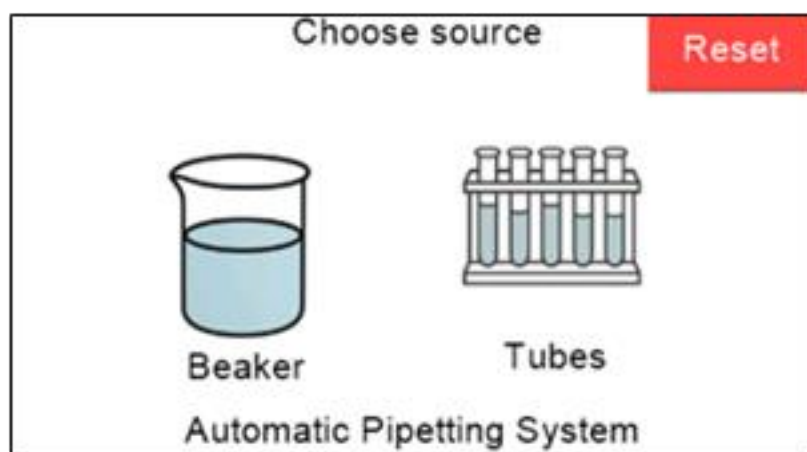
- Tap on the image that corresponds to your liquid source.

Features:

- A Reset button (top-right) restores default settings.

Choose either "Beaker" or "Tubes" to proceed.

Figure H.1: Initial Screen - touch icons on display for different sources (beaker or volume) to take up liquid.



Note: GUI designed using Nextion.

2.2 Page 1a – Beaker Position Selection (only appears if “Beaker” is chosen)

What you see:

- Option to select either Left or Right beaker position (Figure H2).

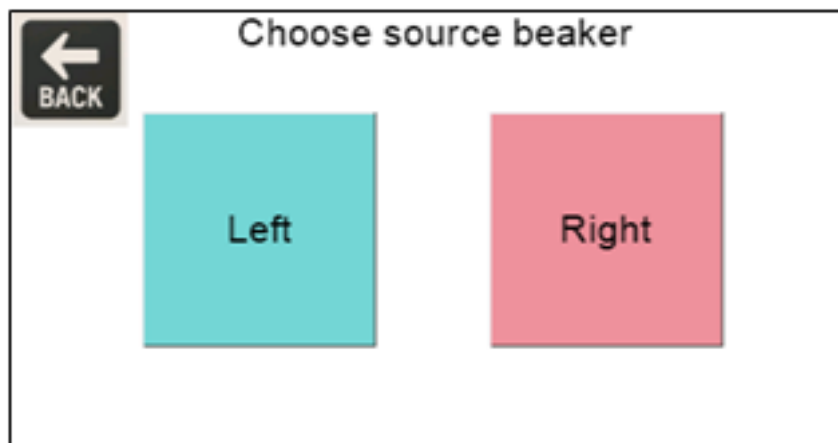
Action:

- Tap the correct side depending on where your source beaker is placed.

Navigation:

- Use the Back button to return to the source type screen if needed.

Figure H.2: Touch icons on display for selecting the left or right beaker.



Note: GUI designed using Nextion. `

2.3 Page 1b – Input Well Selection (only appears if “Tubes” are chosen)

What you see:

- A 5x8 grid representing available wells, with coloured boxes for orientation.

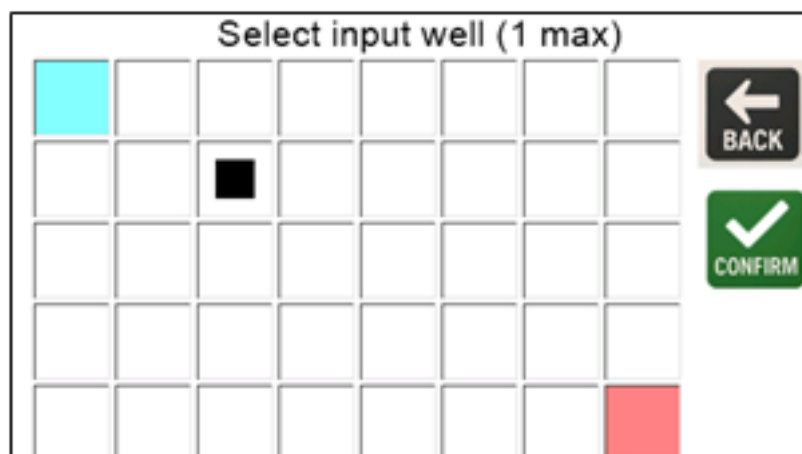
Action:

- Select one input well by tapping a box. Then press Confirm to lock in your choice.

Navigation:

- Use the Back button to return to the source type screen if needed.

Figure H.3: 5x8 Grid displaying source wells



Note: GUI designed using Nextion.

2.4 Page 2 – Output Wells Selection

What you see:

- A second 5×8 grid for selecting one or more output wells.

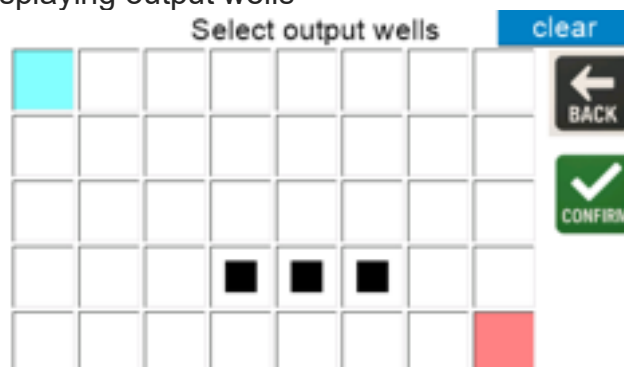
Action:

- Tap the desired wells, then press Confirm to store your choices.

Navigation:

- Clear button resets your selections.
- Back button returns to the previous screen.

Figure H.3: 5x8 Grid displaying output wells



Note: GUI designed using Nextion.

Page 3 – Volume Entry and Final Confirmation

What you see:

- A numeric keypad to enter pipetting volume in microliters (Figure H5).

Action:

- Enter a volume between 10 μ L and 100 μ L.

Navigation:

- Tap Delete to correct any entry errors.
- Tap Start Pipetting to begin the process.

Figure H.5: Numeric keypad displayed on touchscreen to choose desire volume to aspirate/dispense.

Choose volume (uL)

1	2	3
4	5	6
7	8	9
	0	

74 uL






Note: GUI designed using Nextion.

Rules/Errors:

- Only one input well can be selected, and if no well is selected, a warning will appear.
- At least one output well must be selected.
- If a well is selected as input, it cannot also be selected as output, and vice versa. The system will prevent and warn you if this occurs, even when navigating backwards.
- The system checks if the volume is within range, if not, a warning appears and disappears after a short delay.

Appendix I – Gantry Calibration Data

To ensure accurate and reproducible positioning of the pipette head, the gantry system was calibrated along all three axes (X,Y and Z). The goal of this calibration was to establish a reliable conversion between stepper motor pulses and real-world displacement, enabling precise control of pipette's movement during pipetting cycle.

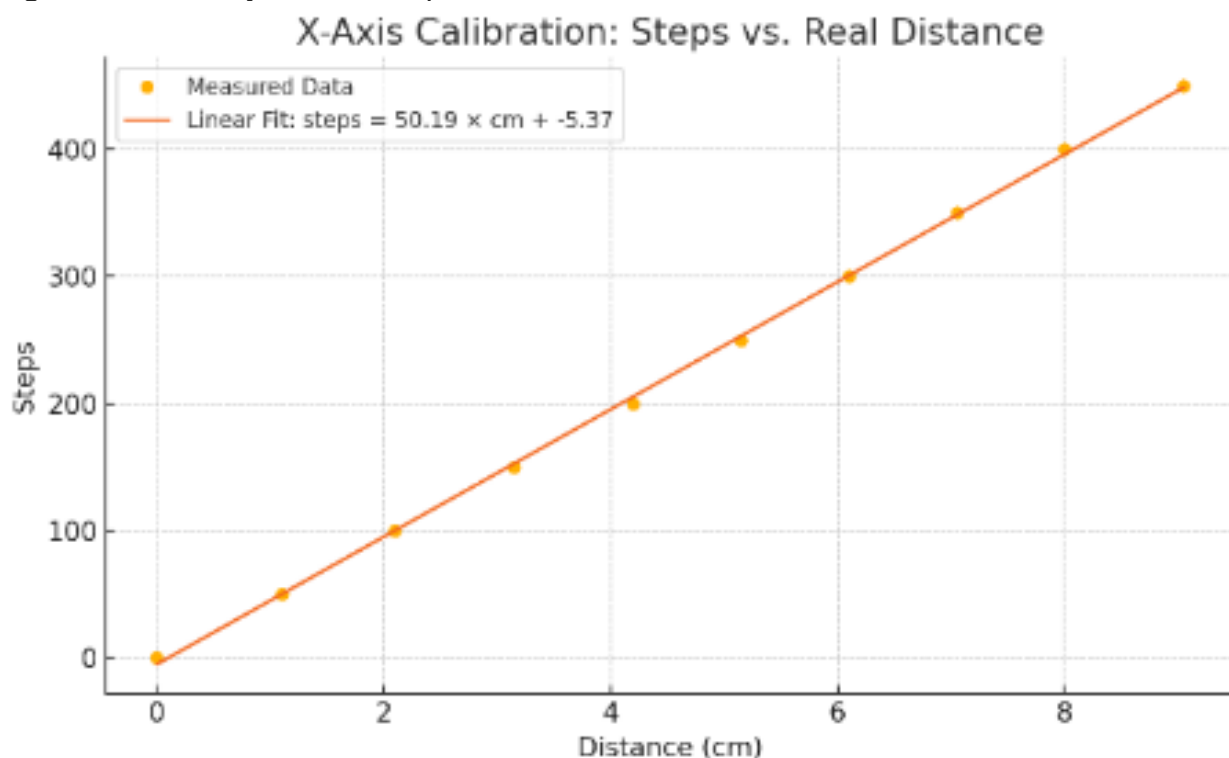
Calibration was carried out by temporarily attaching a pencil to the gantry system in place of a pipette, allowing visible marking of the travel endpoints. Displacements were measured using a ruler at multiple pulse intervals to evaluate both linearity and precision of movement.

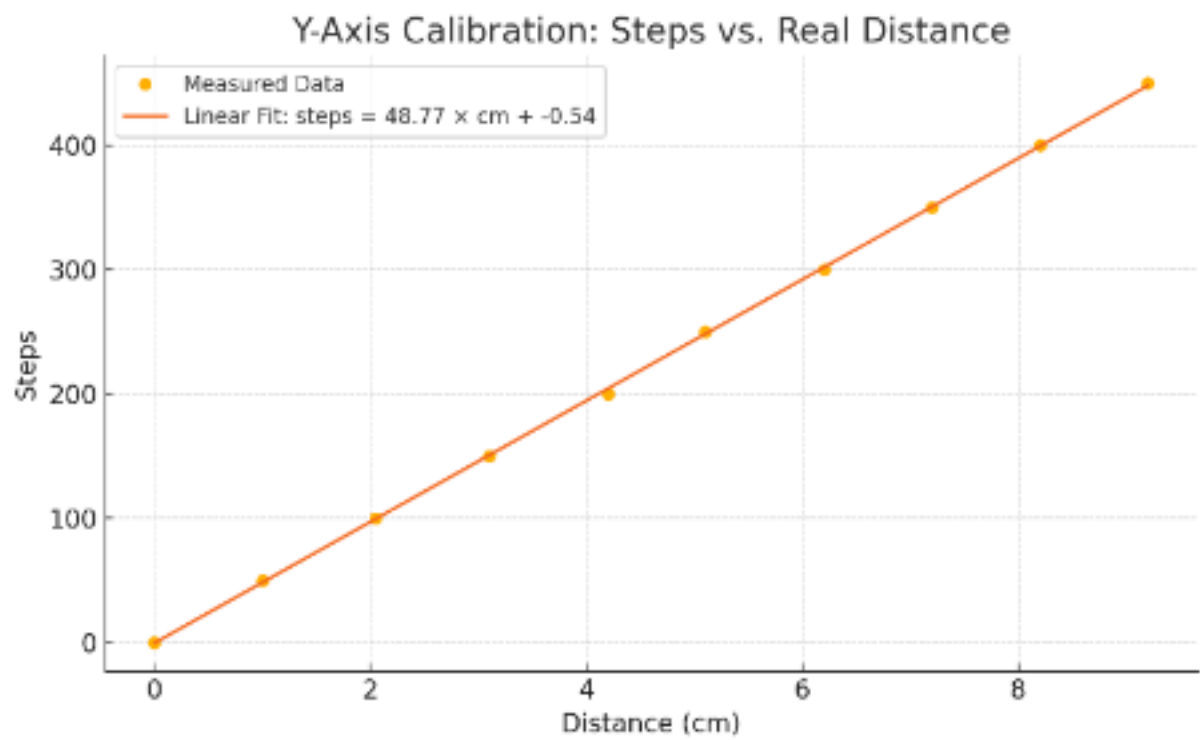
A linear regression method was then implemented to determine the step-to-distance ratio (mm/step) for each axis. The slope of the line of best-fit for each dataset represents the conversion co-efficient which is used to accurately control displacement. The calculated value are shown in *Table G1*, and the relevant plots in *Figure G1*.

Axis	Mm / step
X	0.19
Y	0.20
Z	0.04

Table I.1: Calibrated stepper motor resolution for each axis.

Figure G.1: Gantry calibration plots for X, Y axes





Note: Measured displacement versus step motor pulse counts for each gantry axis. A linear regression was applied to determine the displacement per step. The results show consistent movements in X, Y, Z. These results validate the system's ability to achieve repeatable and accurate positioning across dimensions.

Appendix J – User-Testing Results/Comments

Figure J.1: Response to survey about medical issues

Do you have any past or current medical issues that may limit your physical mobility?
20 responses

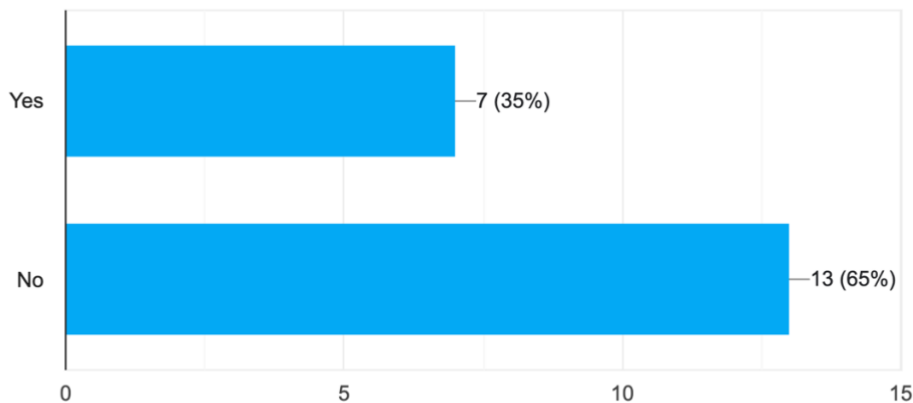


Figure J.2: Portability Satisfaction

Portability: Is the product easy to move around in the lab?
20 responses

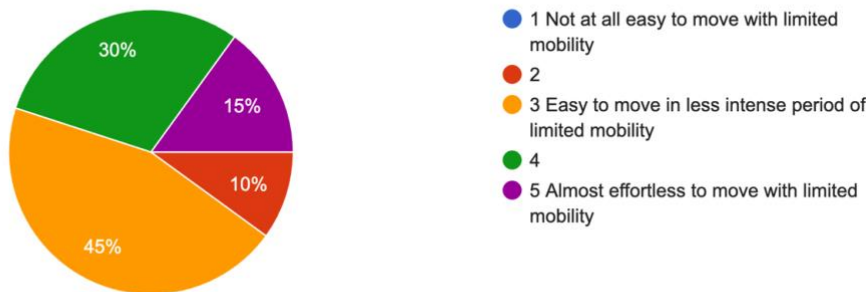


Figure J.3: Aesthetic Satisfaction

Aesthetics: Would the product fit seamlessly into a lab?
20 responses

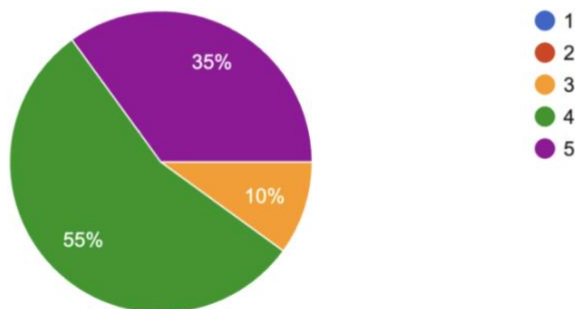


Figure J.4: Feasibiliy

Feasibility: Can you see yourself or someone else regularly using this product in the lab?
20 responses



Figure J.5: Clarity

Clarity: Is the use interface easy to understand/ navigate?
20 responses

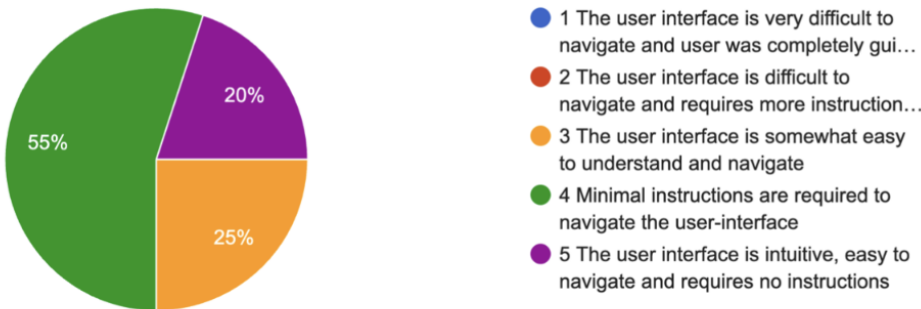


Figure J.6: Ease of use

Pipetting: How easy is it to dispense liquids into glassware or plasticware?
20 responses

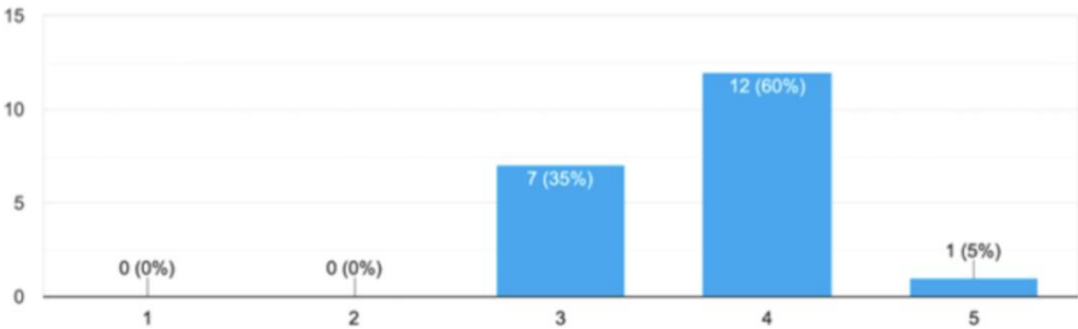


Figure J.7: Improvements

Comments: Any improvements or suggestions for future iterations of the design.

6 responses

Bigger screen
increase in speed or throughput
Tactile feedback would've been preferred
Tactile buttons.
Touchscreen was too small and I struggled to use it
Would've preferred bigger touch icons on the screen